Roma Tre

Information Visualization

Visualization of Time Series

M. Patrignani

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Overview

- Definition of time series data
- Taxonomies of time series data
- Visualization of time series
  - single dependent variable
  - multiple dependent variables
- Examples and case studies

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Definition of time series data

- A data set is called *time series* if
  1. one of the variables is a date or a time
     - each data case may be a registration of an event of some kind
  2. the chronological component is fundamental to the data set
     - the main interest of the user is in the interplay of the other variables with the temporal variable
- Equivalently
  - time is an independent dimension of the data set for which all other variables are given
    - [Müller & Schumann, '03]

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Examples of time series

- Some examples of time series are
  - sunspot activity
  - baseball games
  - medicines taken
  - cities visited
  - stock prices
  - server logs
- 75% of 4000 random samples of graphics from newspapers and magazines from 1974 to 1980 were time series data
  - [Stasko 13] citing [Tufte 83]

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Time series taxonomies

- Taxonomies of time series design
  - based on time axis
    - what are the available data about time?
    - proposed by [Frank ’98]
  - based on dependent variables
    - what other variables are significant?
  - based on time metaphor
    - how time is represented
  - based on user-tasks
    - how time series are used

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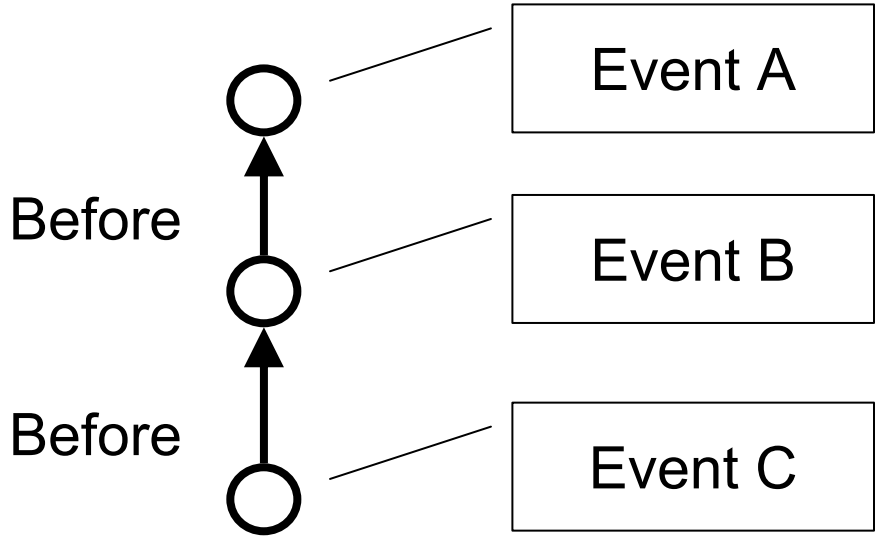
A taxonomy based on time axis

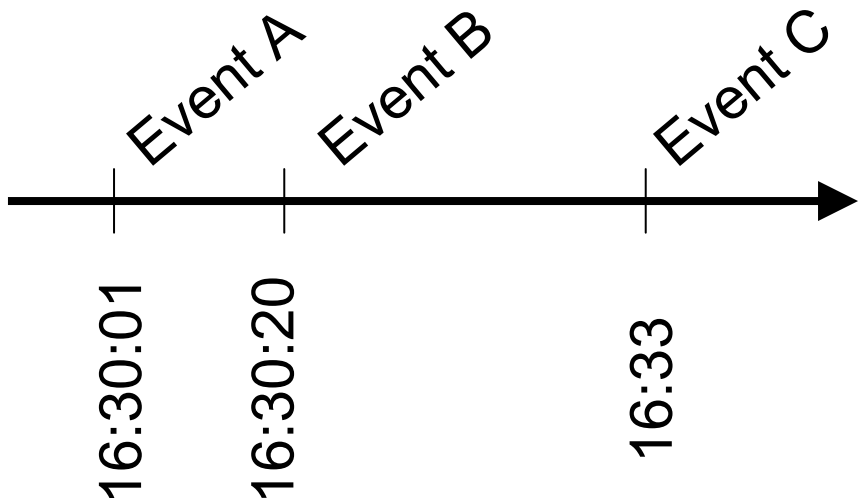
1. Scale
   - ordinal scale versus continuous scale
2. Granularity
   - discrete data versus interval data
3. Arrangement
   - linear versus cyclic time

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1 – Scale

- Ordinal time versus continuous time
  - ordinal time
    - only precedence relationships are known
    - allows for “before” and “after” statements only
  - continuous time
    - allows for quantifying the time difference between any two data points
    - equivalent to say that time is a quantitative variable

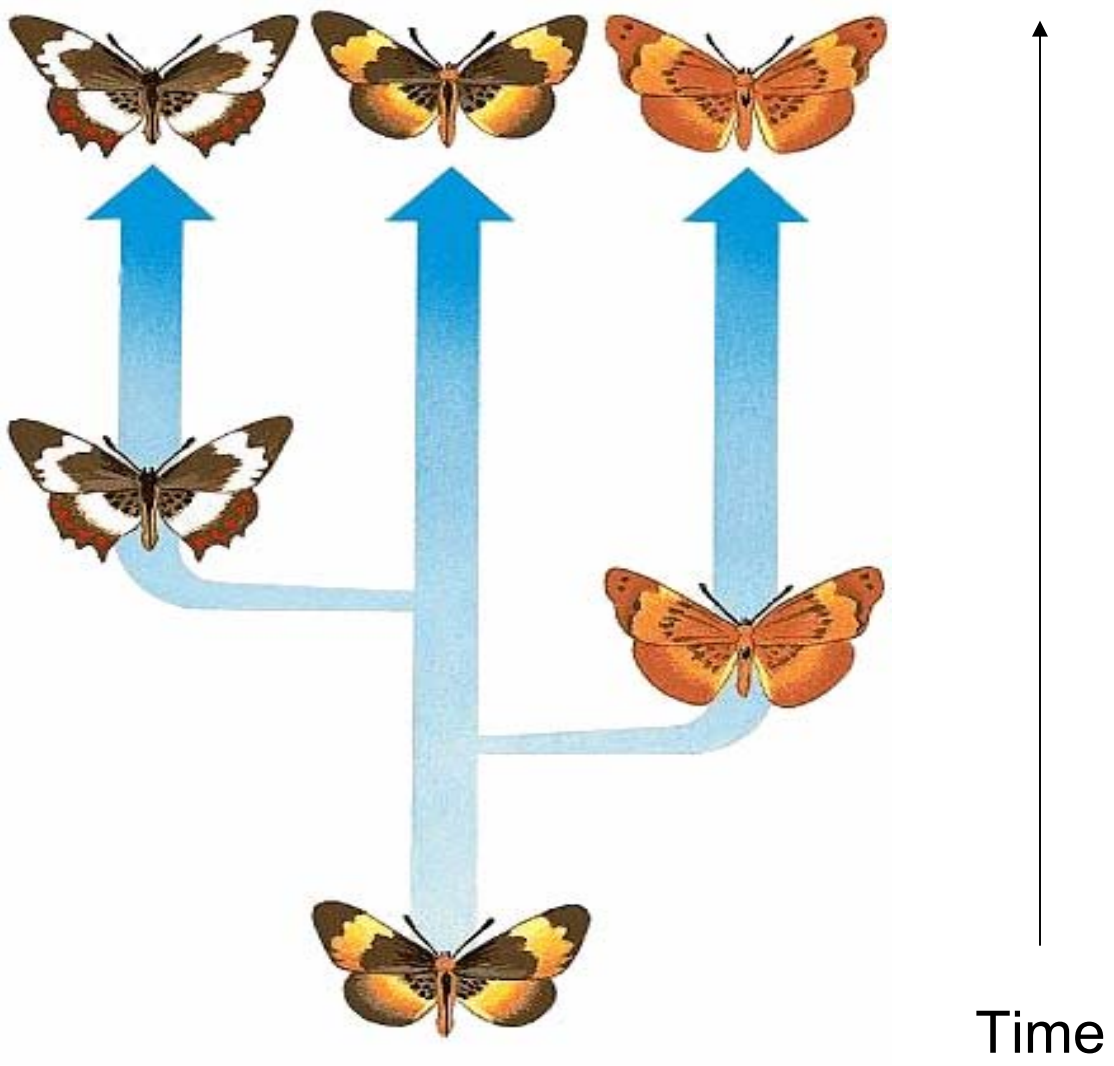




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Example of ordinal scale

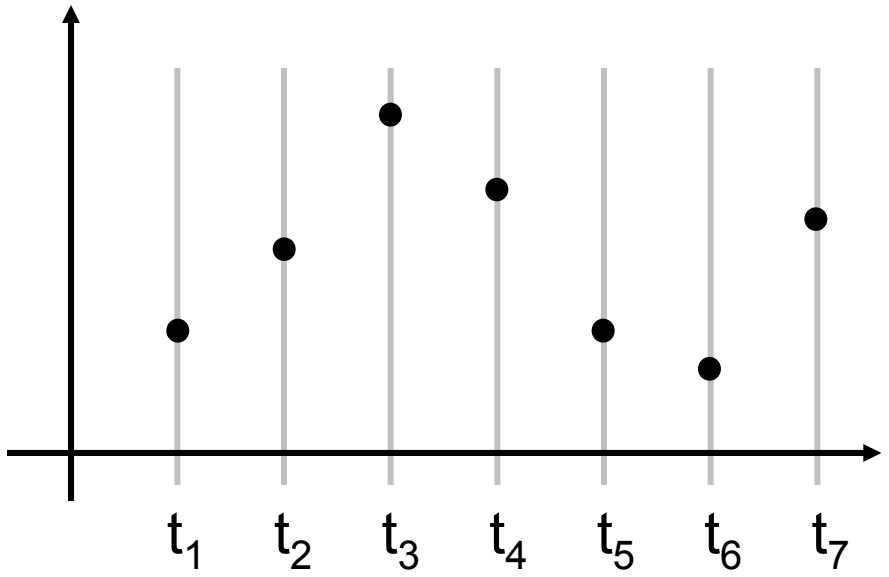
- Evolutionary trees propose ordinal relationships

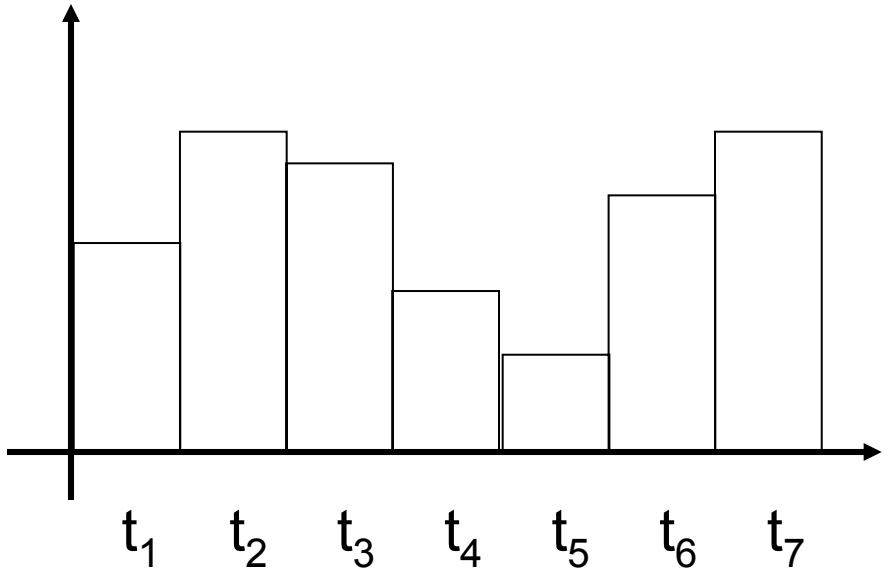


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2 – Granularity

- Discrete points versus interval points
  - discrete points are instantaneous values
  - interval points are values associated to intervals

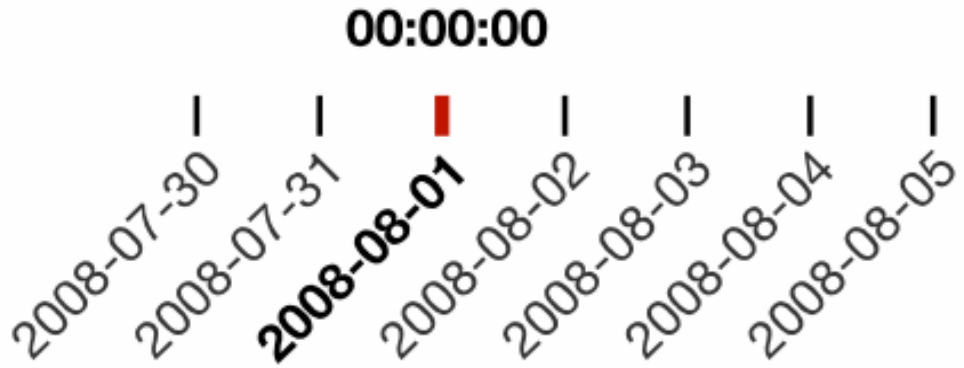


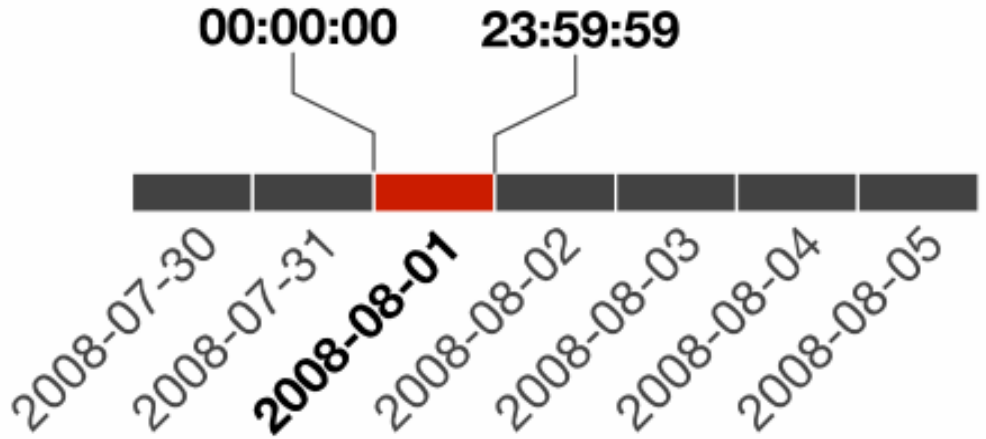


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Granularity: examples

- Time value “August 1, 2008” in a point-based domain
  - no information is given in between two time points
- Time value “August 1, 2008” in an interval-based domain
  - each element covers a subsection of the time domain greater than zero





[Pictures and text from [Aigner et al. 11]

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Granularity: examples

- An event log generally offers discrete points
- Aggregate values are usually referred to time intervals

```
root@tittolo:/var/log# tail auth.log
Apr 15 17:09:01 CRON[13984]: session opened for user root by (uid=0)
Apr 15 17:09:01 CRON[13984]: session closed for user root
Apr 15 17:17:01 CRON[14102]: session opened for user root by (uid=0)
Apr 15 17:17:02 CRON[14102]: session closed for user root
Apr 15 17:39:01 CRON[14183]: session opened for user root by (uid=0)
Apr 15 17:39:01 CRON[14183]: session closed for user root
Apr 15 18:05:11 sudo: USER=root ; COMMAND=/bin/bash
Apr 15 18:05:11 sudo: session opened for user root by patrigna(uid=0)
Apr 15 18:09:01 CRON[14363]: session opened for user root by (uid=0)
Apr 15 18:09:01 CRON[14363]: session closed for user root
root@tittolo:/var/log#
```

Ora	Tempo	T (°C)	Vento (km/h)	Precipitazioni
02.00	poco nuvoloso	12.2°	SW 4 / max 6 debole	<0.05 mm isolate
05.00	sereno	10.7°	SW 5 / max 8 debole	- assenti -
08.00	poco nuvoloso	13.7°	W 2 debole	0.2 mm deboli
11.00	pioggia e schiarite	17.5°	W 5 / max 6 debole	0.4 mm deboli
14.00	pioggia e schiarite	19.6°	WSW 12 / max 13 moderato	0.4 mm deboli
17.00	pioggia e schiarite	19.4°	WSW 16 / max 18 moderato	0.3 mm deboli
20.00	pioggia e schiarite	15.2°	WSW 10 / max 15 debole	0.5 mm deboli
23.00	pioggia e schiarite	13.4°	SW 5 / max 8 debole	0.6 mm deboli
02.00	poco nuvoloso	11.8°	calma	0.3 mm deboli

From www.ilmeteo.it

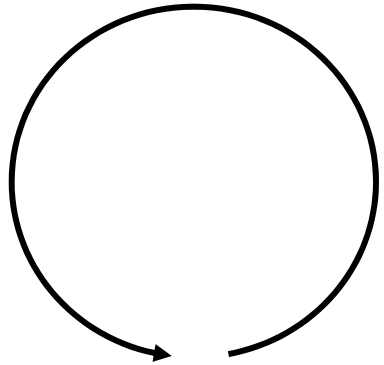
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3 – Arrangement

- Linear time versus cyclic time
- Two ways of arranging the visualizzation
  - linear time is represented as continously flowing in one direction
  - cyclic time is represented has having an intrinsic periodicity



Linear time



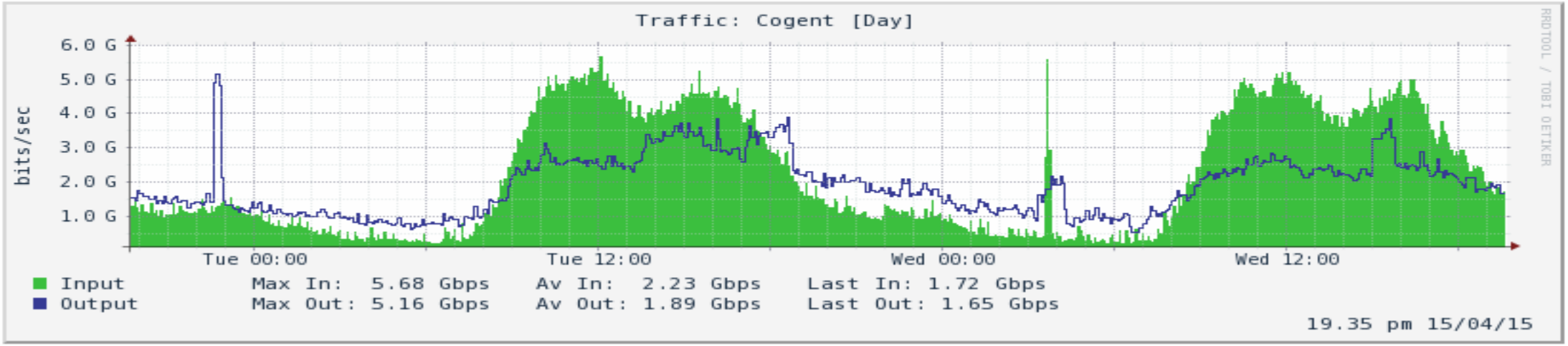
Cyclic time

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Linear time example

- Possible periodicity has to be evinced by the user

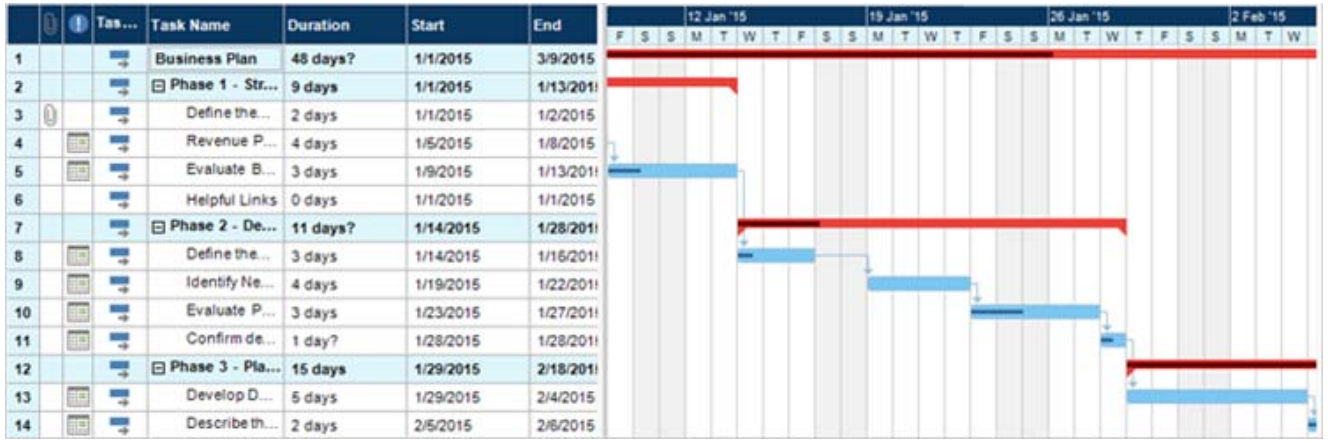
Link Name	Link Type	Equipment	IP Address	Interface	Url
Cogent - PoP RM2	Direct 10Gbps 10GE	r.rm2.garr.net	149.6.22.74	xe-3/1/0.0	



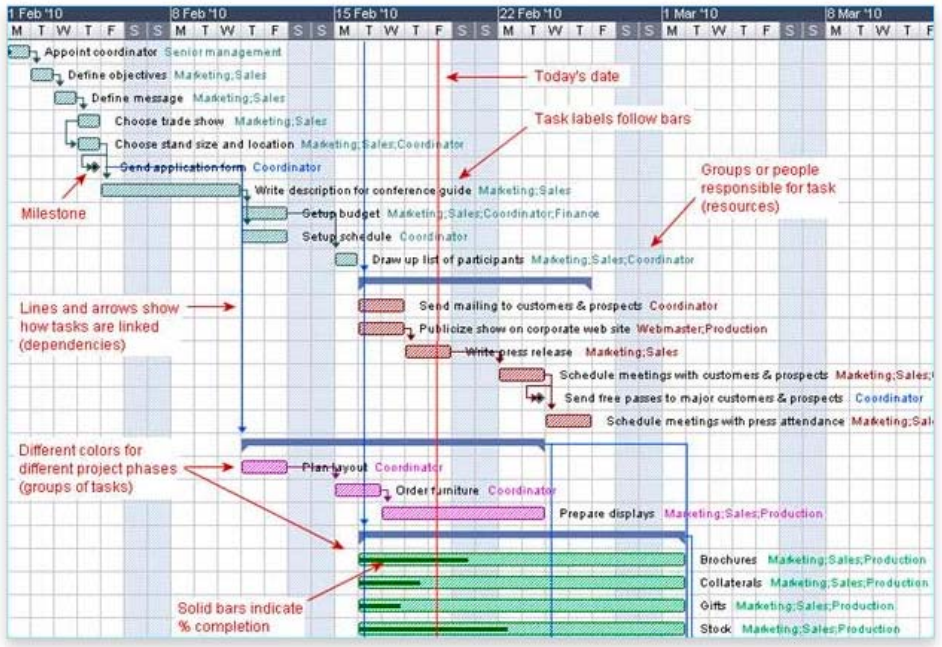
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Linear time example: Gantt chart

- Devised in the mid 1890s by the Polish engineer Karol Adamiecki
- Popularized in 1900 by the American engineer Henry Gantt



	Task Name	Duration	Start	End
1	Business Plan	48 days?	1/1/2015	3/9/2015
2	Phase 1 - Str...	9 days	1/1/2015	1/13/201
3	Define the...	2 days	1/1/2015	1/2/2015
4	Revenue P...	4 days	1/5/2015	1/8/2015
5	Evaluate B...	3 days	1/9/2015	1/13/201
6	Helpful Links	0 days	1/1/2015	1/1/2015
7	Phase 2 - De...	11 days?	1/14/2015	1/28/201
8	Define the...	3 days	1/14/2015	1/16/201
9	Identify Ne...	4 days	1/19/2015	1/22/201
10	Evaluate P...	3 days	1/23/2015	1/27/201
11	Confirm de...	1 day?	1/28/2015	1/28/201
12	Phase 3 - Pla...	15 days	1/29/2015	2/18/201
13	Develop D...	5 days	1/29/2015	2/4/2015
14	Describe th...	2 days	2/5/2015	2/6/2015

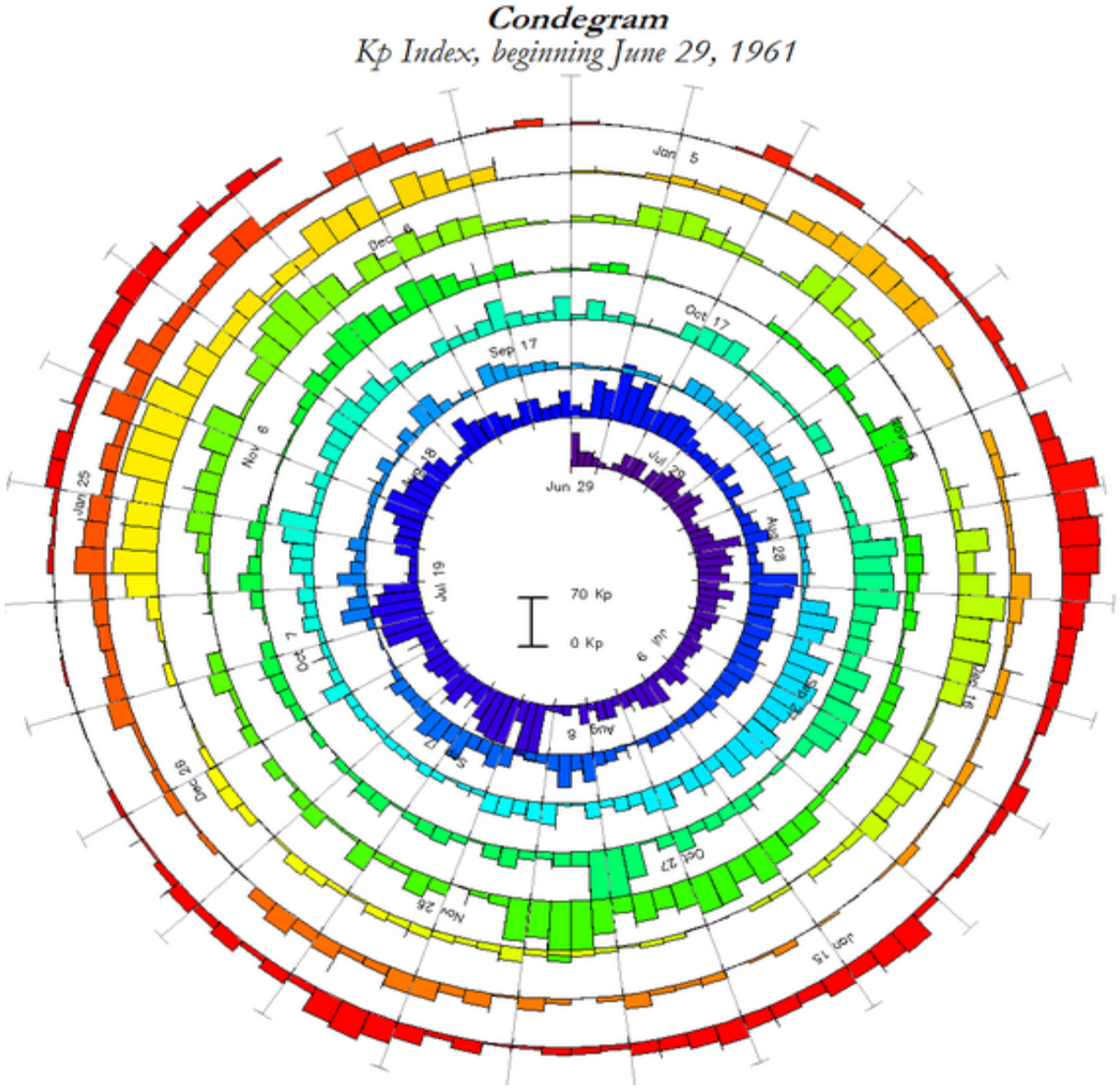


From www.gantt.com

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Cyclic time example

- Used for datasets that exhibit an inherent periodical pattern
  - Callum Anderson, University of Alaska Fairbanks



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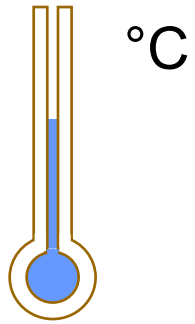
Taxonomy based on variables

- Time is the only independent variable
- Single dependent variable
  - examples
    - inbound/outboud traffic on a network link
    - power plant electricity usage
    - pollution levels
    - …
- Multiple dependent variables
  - examples
    - stock market price of shares
    - weather forecasts (temperature, humidity, precipitations, …)
    - …

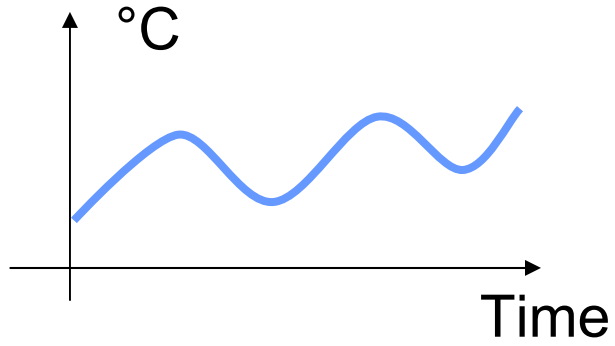
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Taxonomy based on time metaphor

- Time is mapped to time
  - easiest choice
  - dynamic (time-dependent, evolving, animated) visualization
  - needs continous attention, but does not use much space



- Time is mapped to space
  - more difficult choice
  - static visualization
  - does not rely on memory, allows comparisons, needs more space



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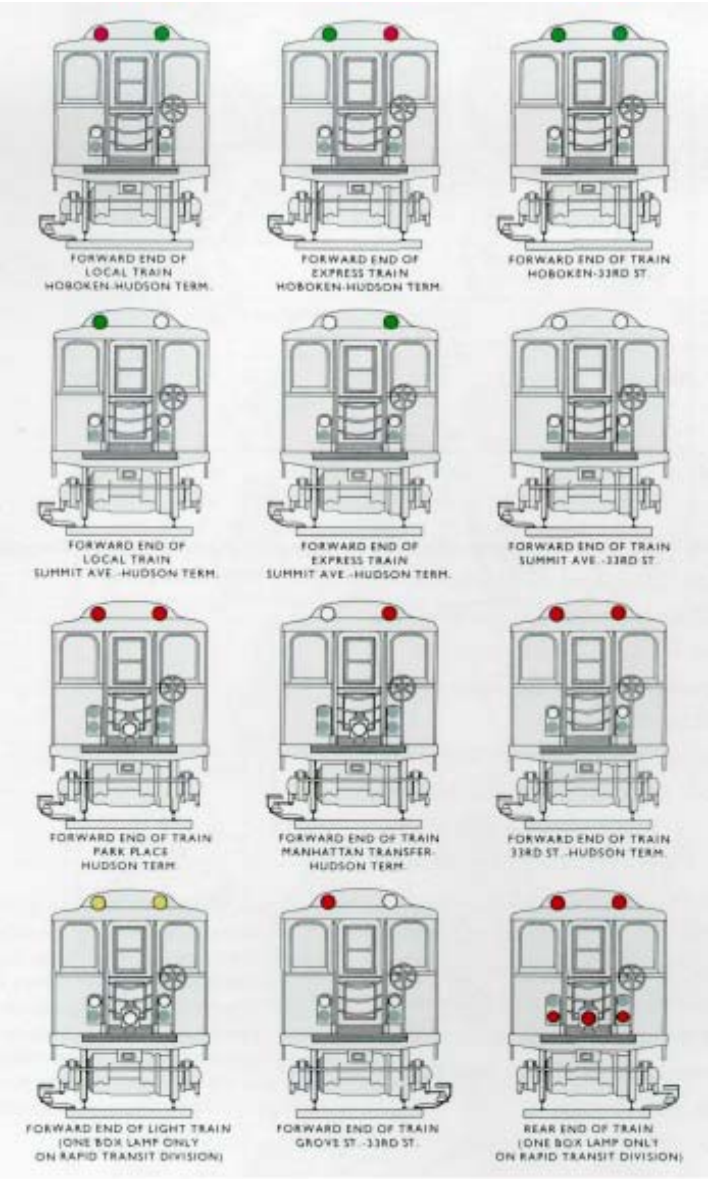
Direct comparison Vs memory

- Direct comparison = all information is represented at the same time
  - examples
    - small multiples, multiple bars, multiple views
- Memory = information is available on a need-to-know basis
  - the different views are available one at a time
  - interaction to browse among views
    - for example with a timeline or with animation controls

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Direct comparison Vs memory

- Eyes beat memory
  - using our eyes to switch between multiple views has less cognitive load than consulting our memory to compare current and previous views
    - From [Munzner 14]
- Using space is more effective than using time and relying on memory



Picture from [Turfe 90]
Hudson & Manhattan Railroad Company
New York, 1923

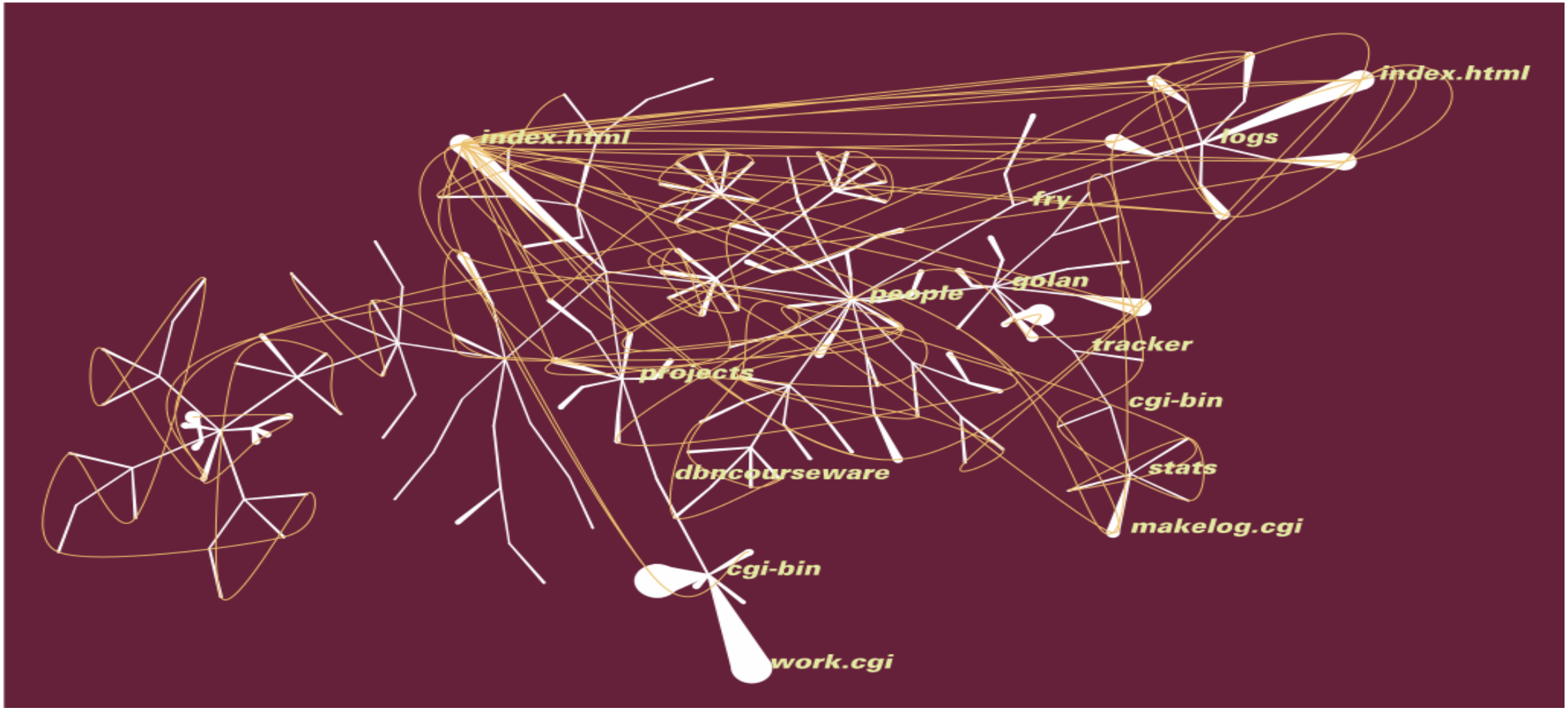
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Taxonomy based on user tasks

- Point oriented
  - focus on the values at specific times
  - check and compare instantaneous values
- Time oriented
  - search for specific periods of time
  - find intervals with high/low values
- Trend oriented
  - discover patterns, trends, interplay
  - compare behavior
  - find regularity or periodicity

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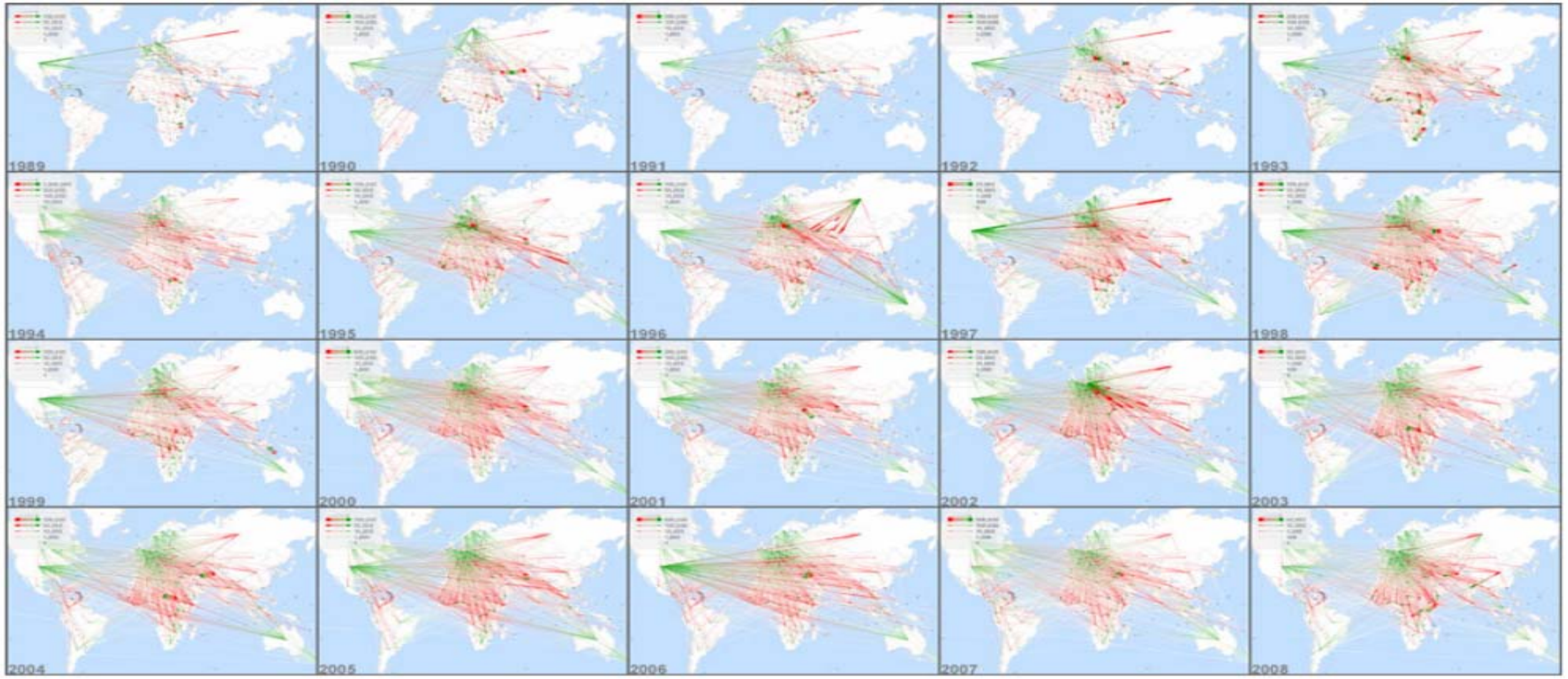
Exploiting animation: Anemone



- Dynamically shows traffic patterns of people visiting a web site
  - [Fry 00]

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Small Multiples

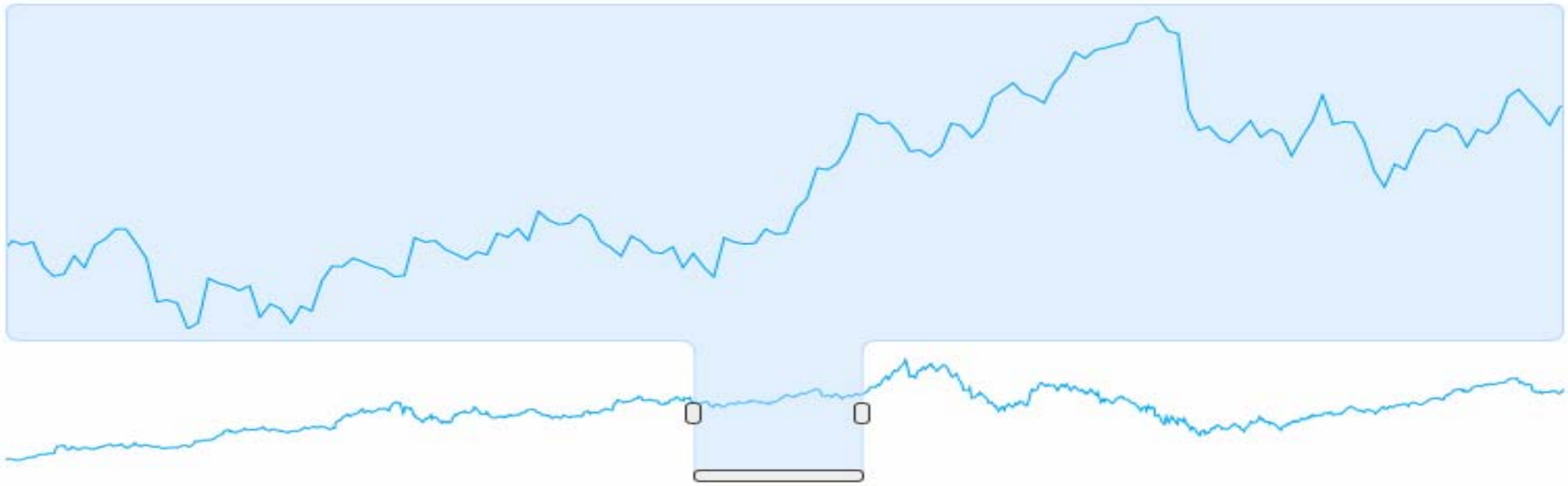


- Migration data: red color indicates the origin, green color the destination
  - generated with JFlowMap by Ilya Boyandin [Aigner *et al.* 11]

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Exploration of a line graph

- Envision.js TimeSeries
  - zoom support for exploring details while maintaining the context

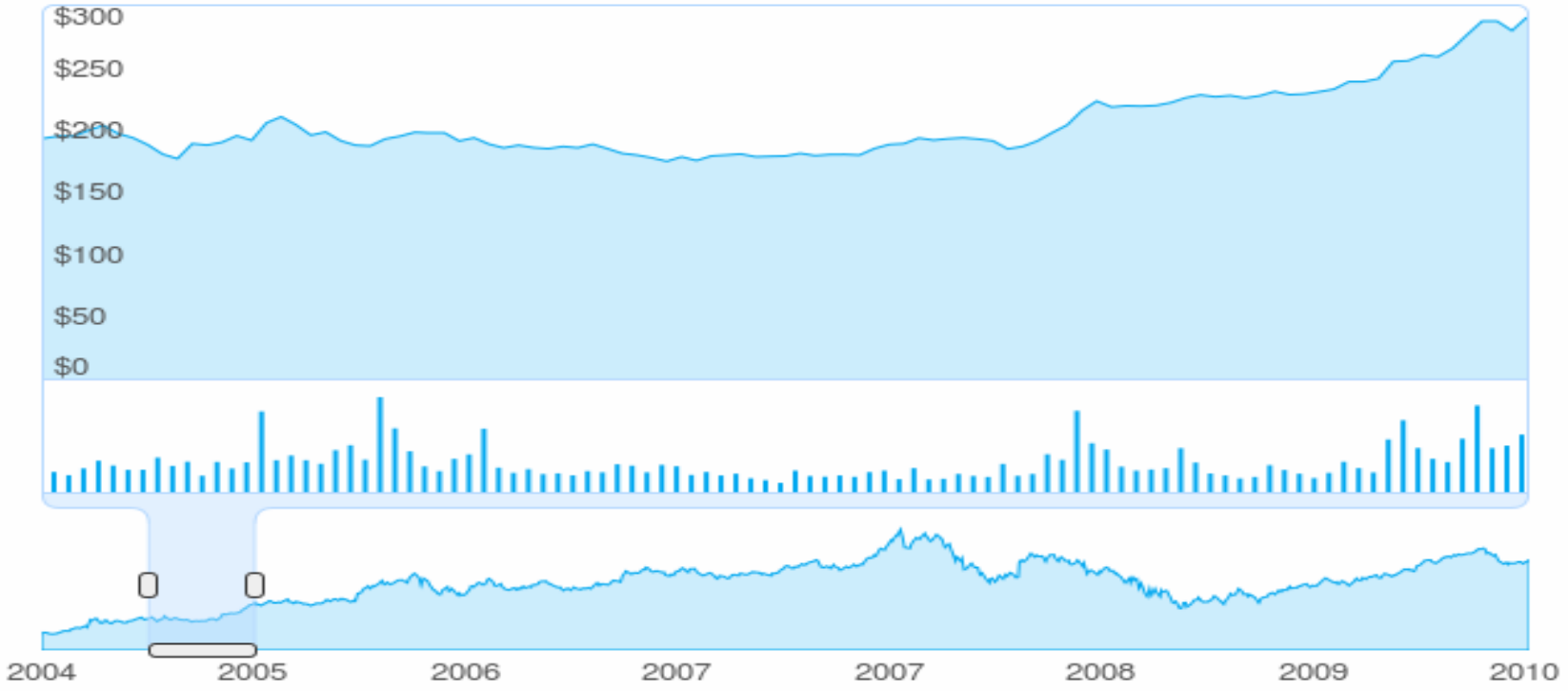


http://www.humblesoftware.com/envision/demos/timeseries

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Visualization of two variables

- Envision.js Finance
  - zoom support for the visualization of price and quantity exchanged of a stock

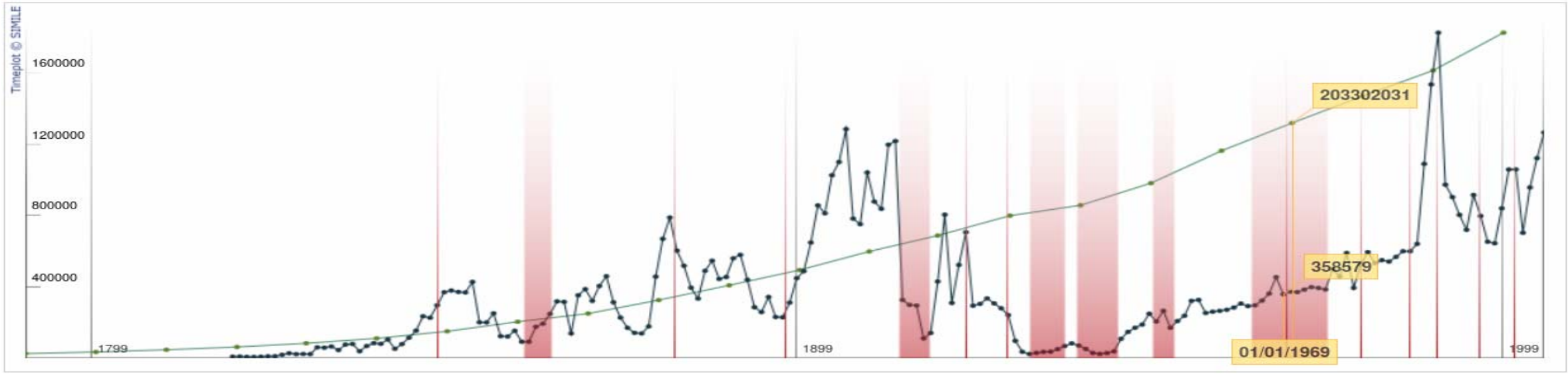


http://www.humblesoftware.com/envision/demos/finance

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Visualization of two variables

- SIMILE timeplot
  - open source widget
  - visualization of U.S. population, U.S. immigration and highlighted periods

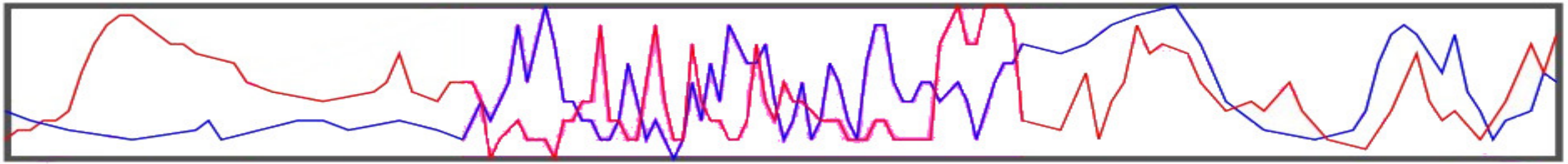


http://simile-widgets.org/timeplot/

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Stack zooming

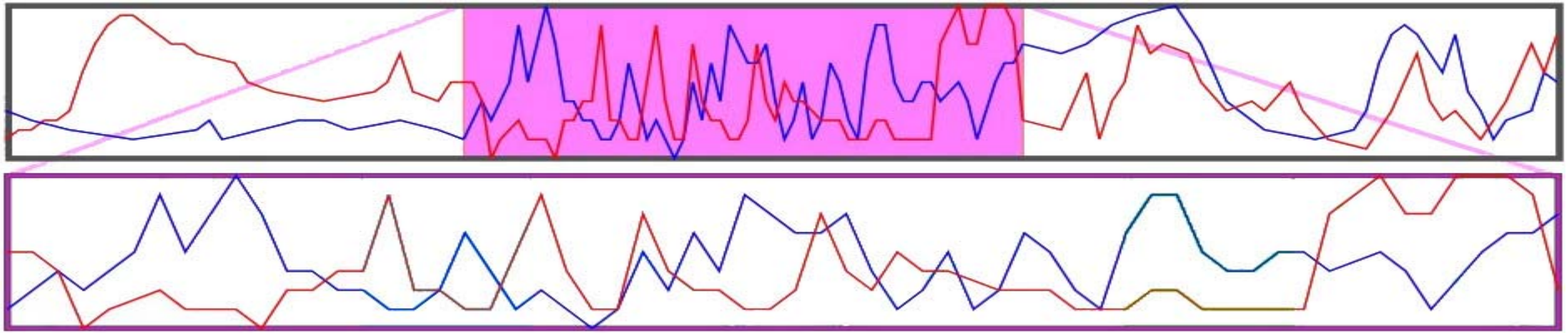
- multi-focus interaction in temporal data
  - from [Javed & Elmqvist, 10]



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Stack zooming

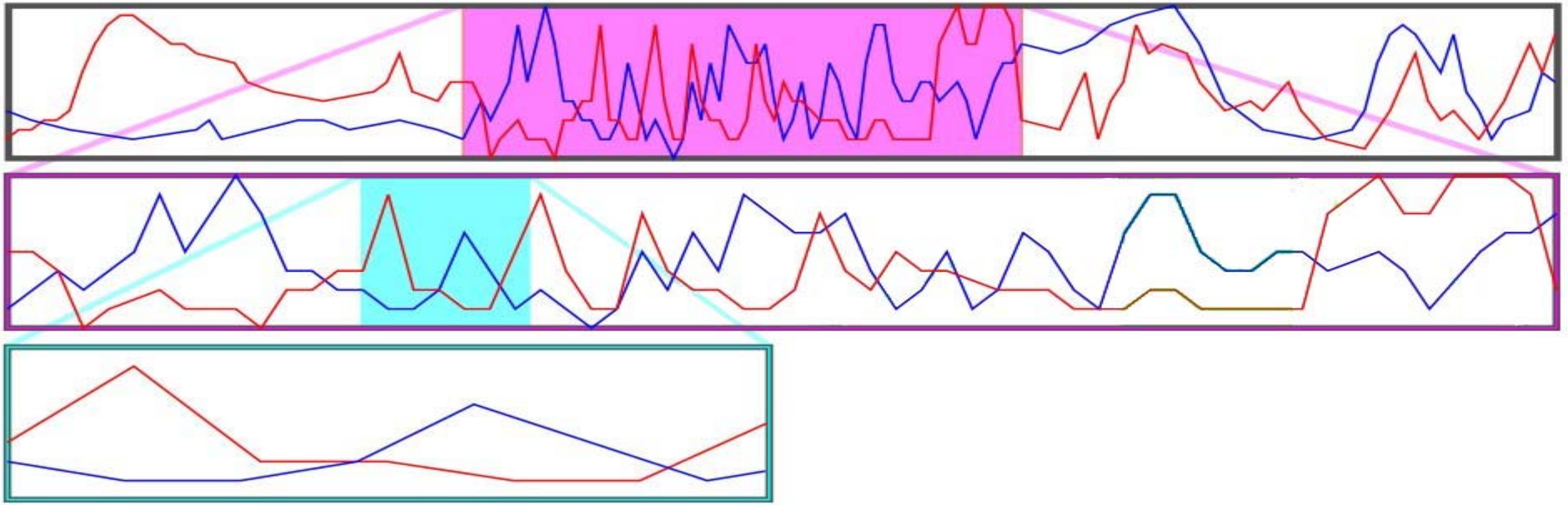
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Stack zooming

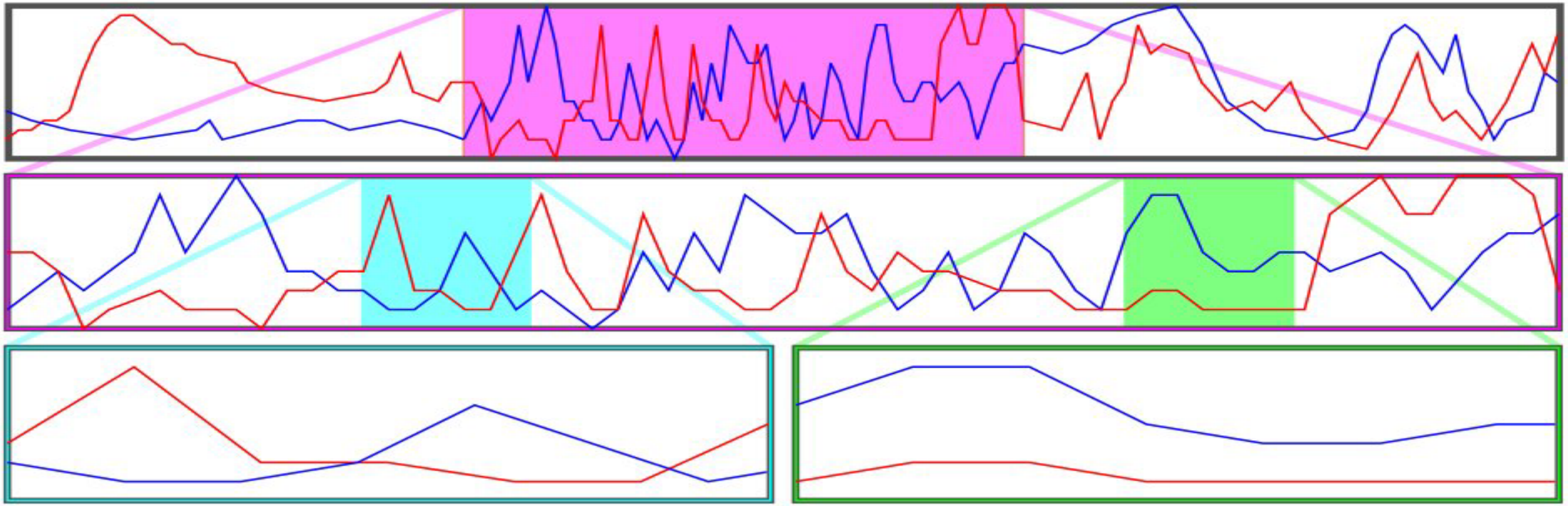
- multi-focus interaction in temporal data
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Stack zooming

- multi-focus interaction in temporal data
  - from [Javed & Elmqvist, 10]



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The TraXplorer system

- From [Javed & Elmqvist, 10]



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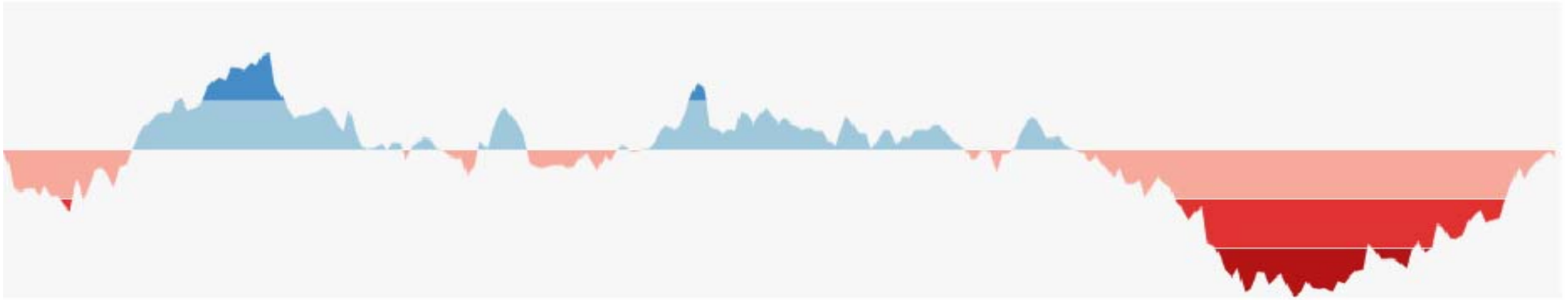
Horizon graphs



1. Start with a regular line graph
   - centered at its baseline

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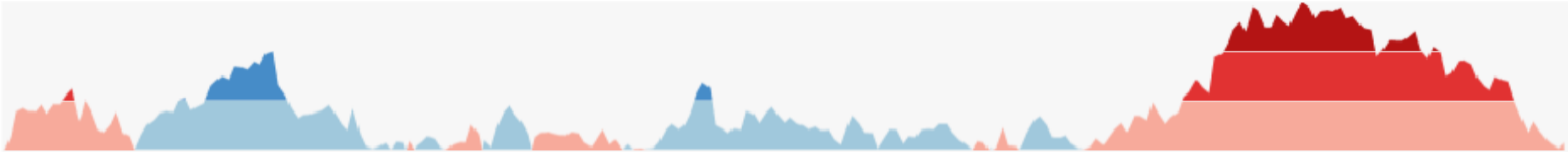
Horizon graphs



1. Start with a regular line graph
   - centered at its baseline
2. Color it with layers
   - blue is positive, red is negative

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Horizon graphs



1. Start with a regular line graph
   - centered at its baseline
2. Color it with layers
   - blue is positive, red is negative
3. Mirror around its baseline

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Horizon graphs



1. Start with a regular line graph
   - centered at its baseline
2. Color it with layers
   - blue is positive, red is negative
3. Mirror around its baseline
4. Collapse bands into a single one

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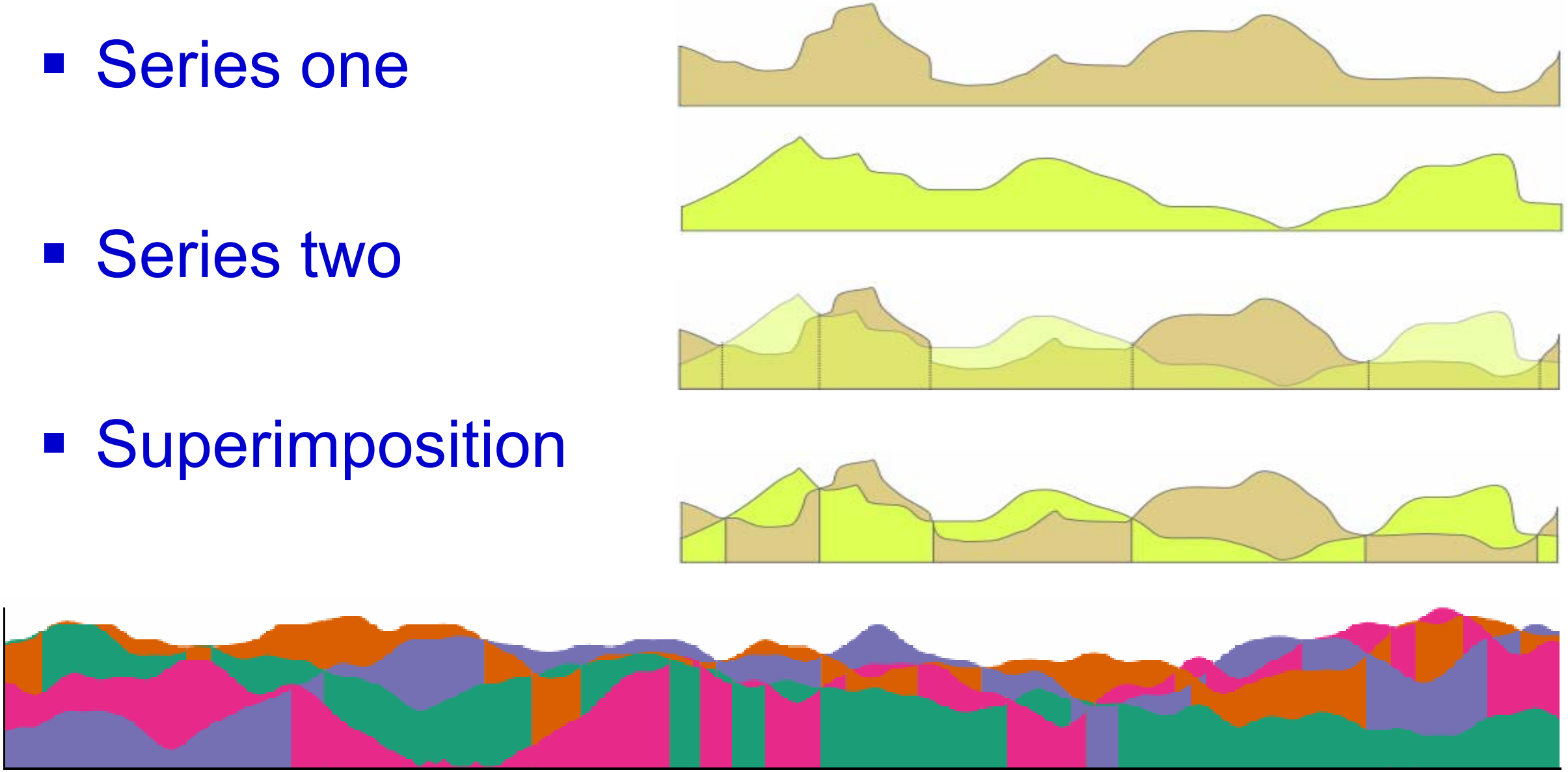
Horizon graphs



- Introduced by [Reijner 08]
- Why “horizon graphs”?

Braided graphs

- Series one
- Series two
- Superimposition

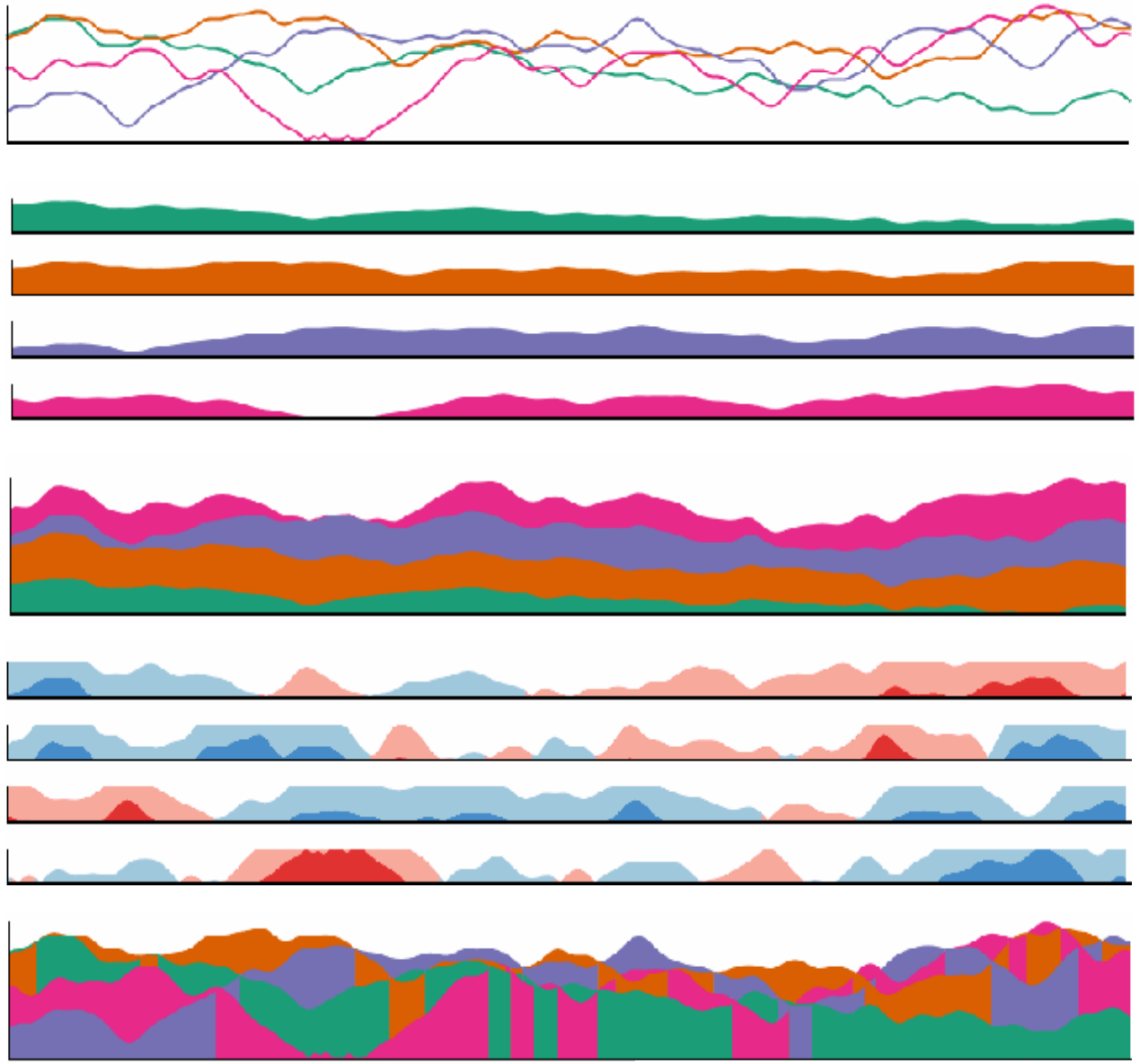


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[Javed *et al.*, '10]

Multiple time series visualizations

- Line graph
- Small multiples filled line graphs
- Stacked graph
- Small multiples horizon graphs
- Braided graph

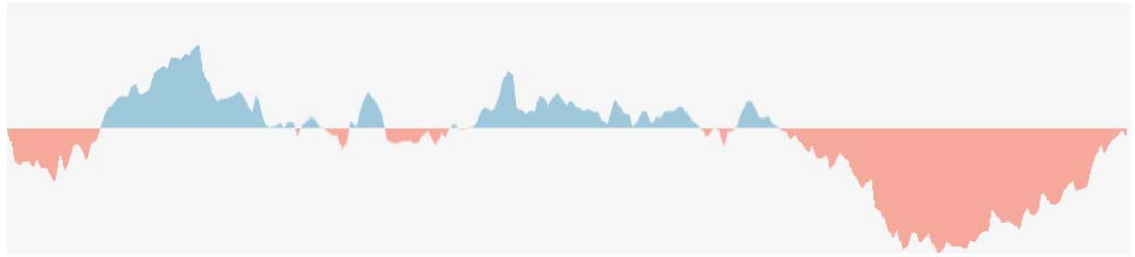


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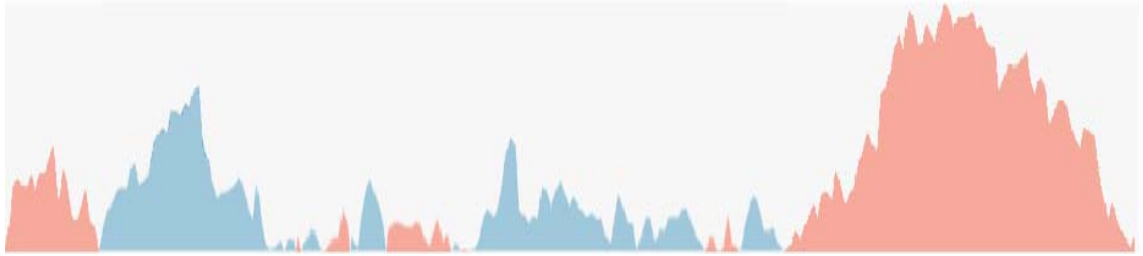
[Javed *et al.*, '10]

Evaluation: horizon graphs

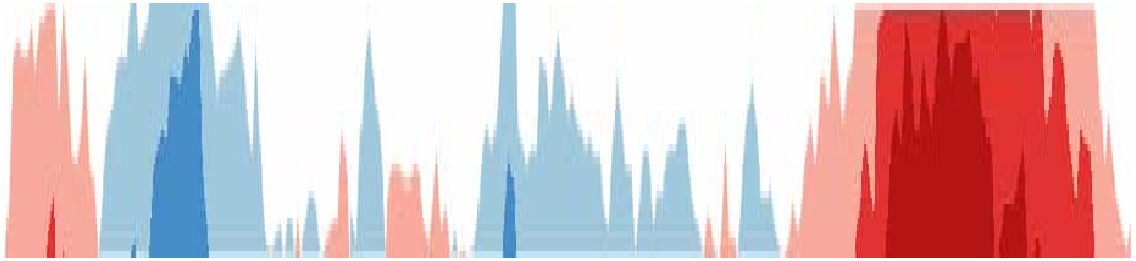
- From [Heer *et al*., ‘09]
  - mirroring does not slow estimation nor hurt accuracy
    - always advisable
  - horizon graphs are beneficial if their height is very small
    - threshold is 6.8 mm or 24 pixels
    - Horizon graphs with three overlaps is undecided
    - Horizon graphs with four overlaps is discouraged



Filled line graph



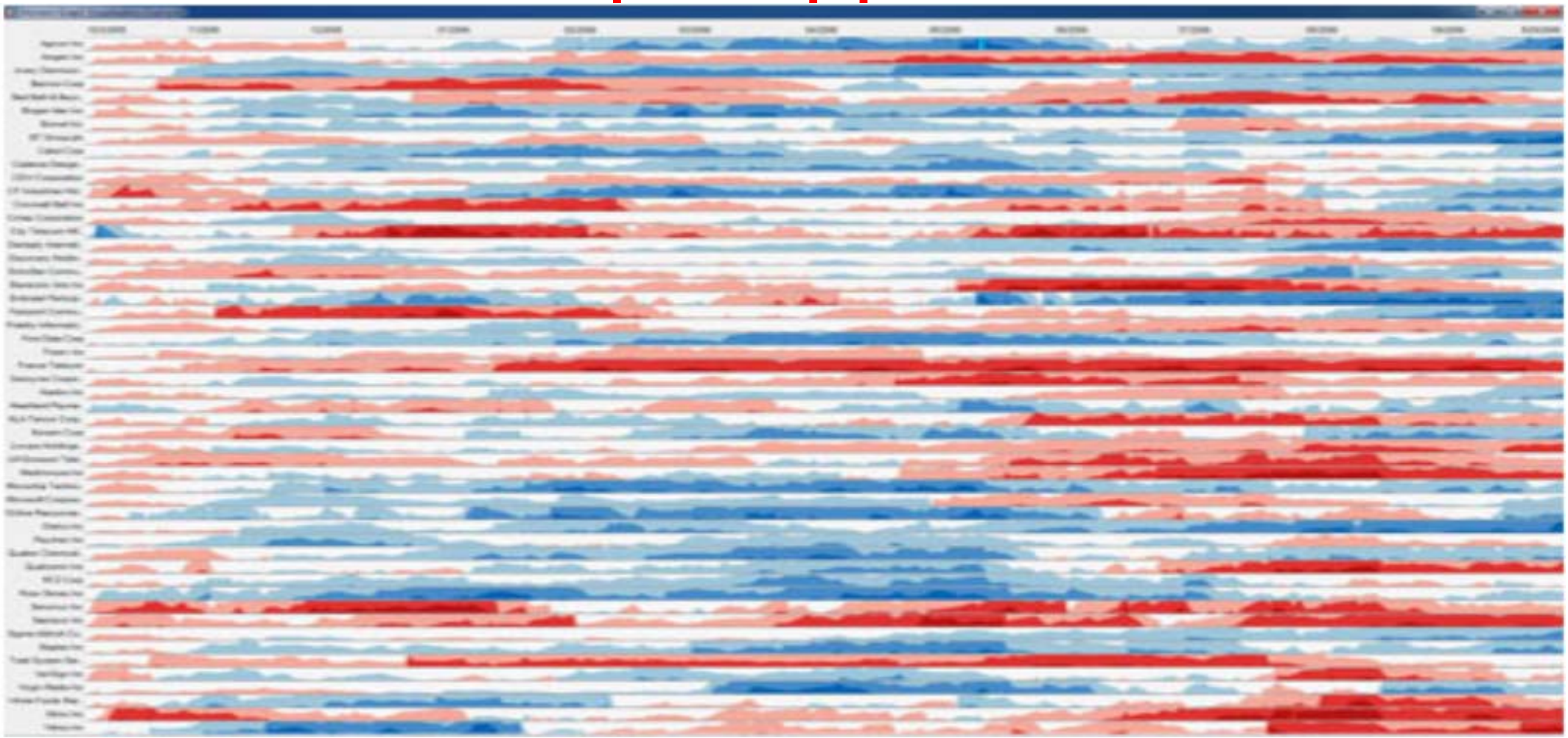
Mirrored graph



Horizon graph (three overlaps)

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Example application

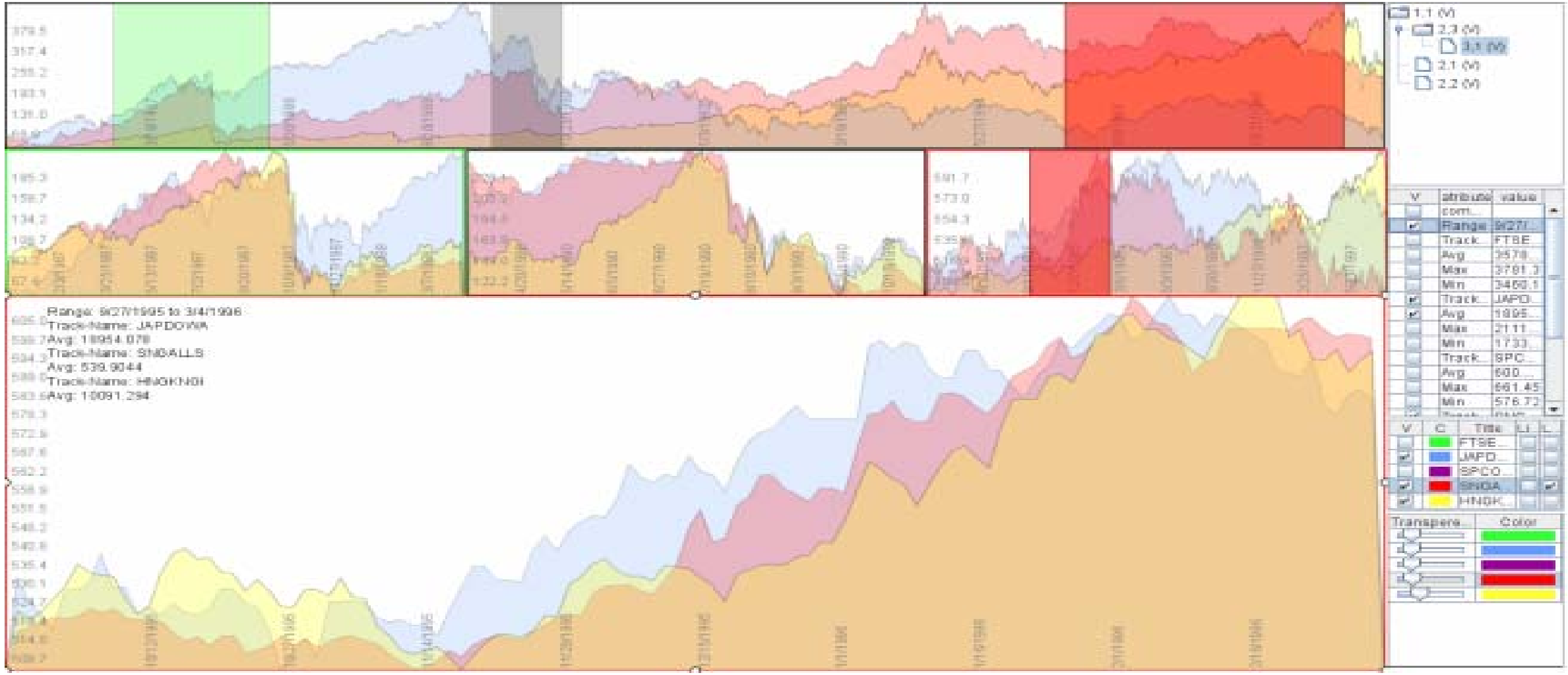


- Comparing stock market data
  - picture from [Aigner *et al.* 11], courtesy of Hannes Reijner [Reijner 08]

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TraXplorer with filled line graphs

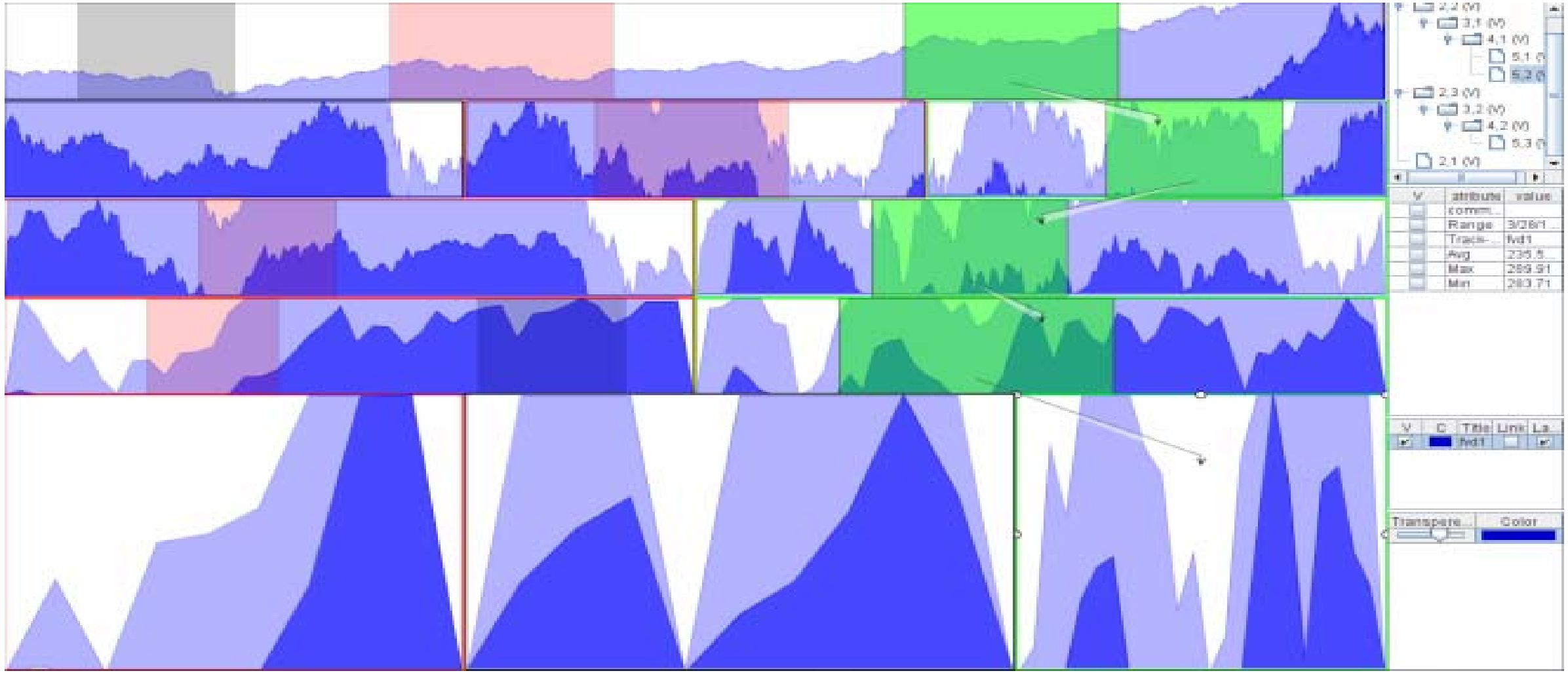
- From [Javed & Elmqvist, 10]



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TraXplorer with horizon graphs

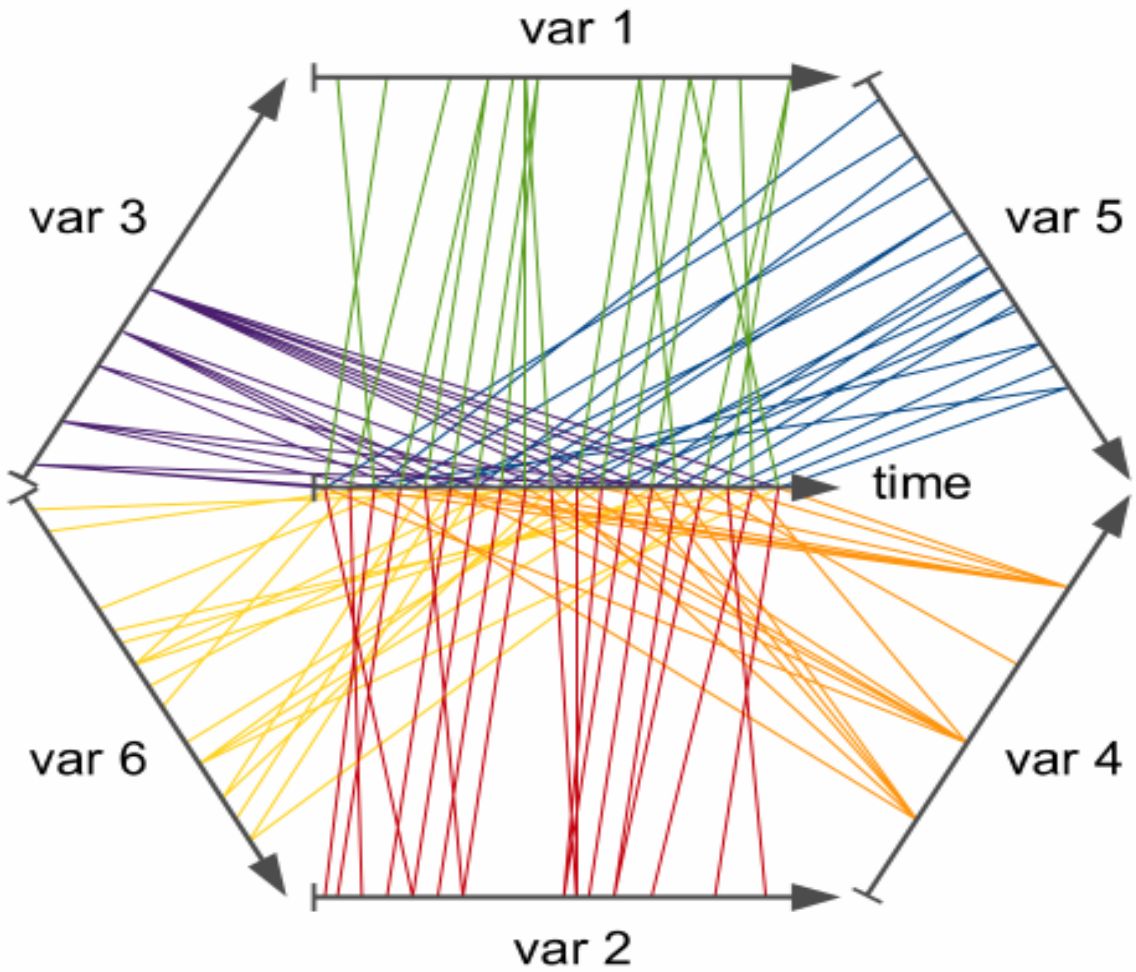
- From [Javed & Elmqvist, 10]



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TimeWheel

- Variable axes are arranged around the horizontal point-based time axis in a regular polygonal manner
- For each time-step, lines descend from the time axis to the corresponding points on the variable axes
  - introduced by [Tominski et al 04]

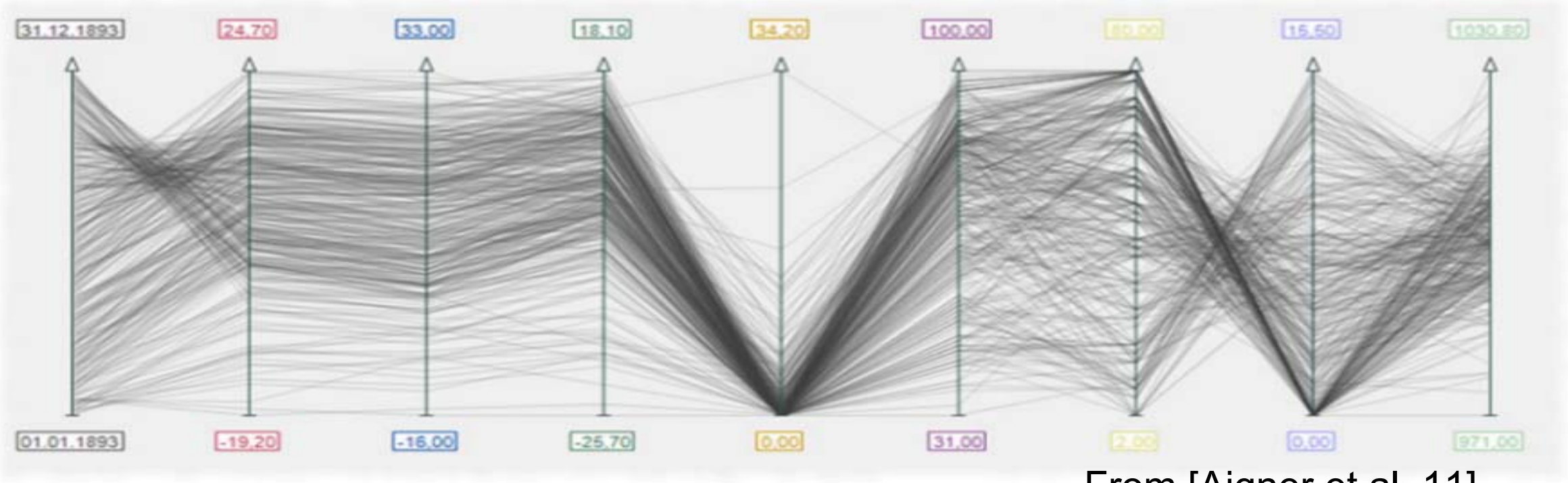


Pictures and text from [Aigner et al. 11]

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Parallel coordinates vs timewheels

- In parallel coordinates the time axis (here the leftmost axis) is just one of many axes and it is not treated in any particular way to emphasize the importance of time



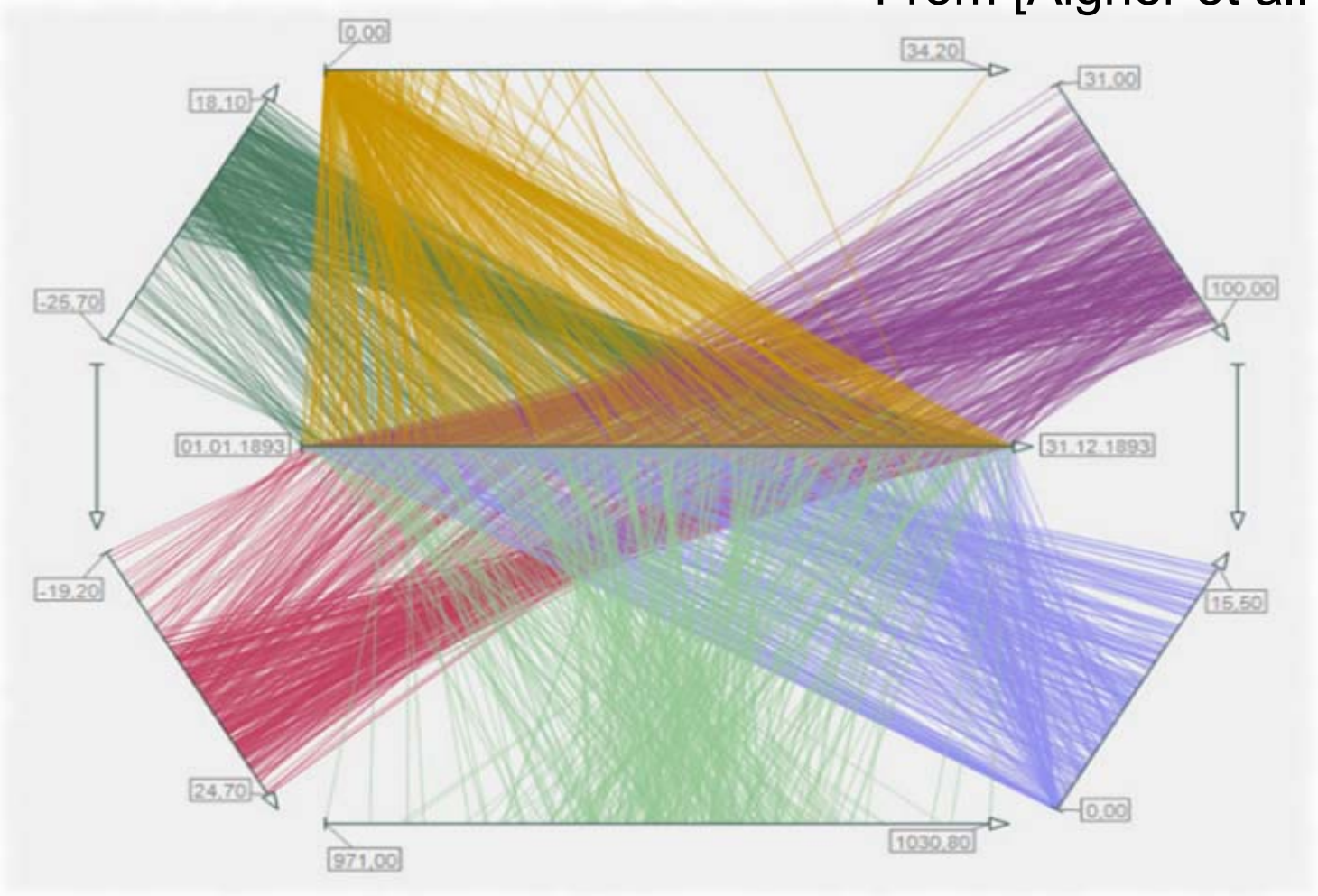
From [Aigner et al. 11]

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Parallel coordinates vs timewheels

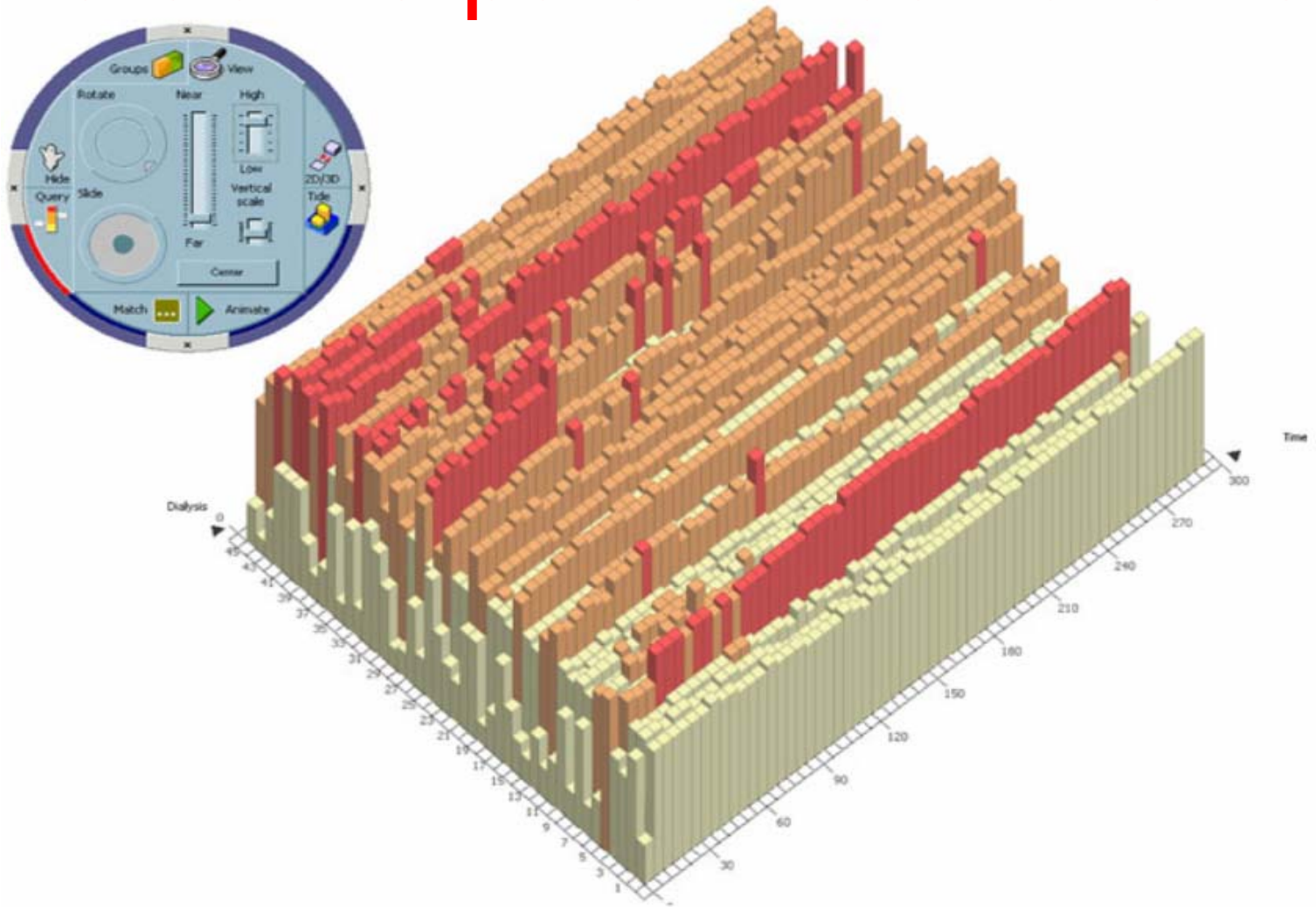
- Reference time axis is in a prominent central position

From [Aigner et al. 11]



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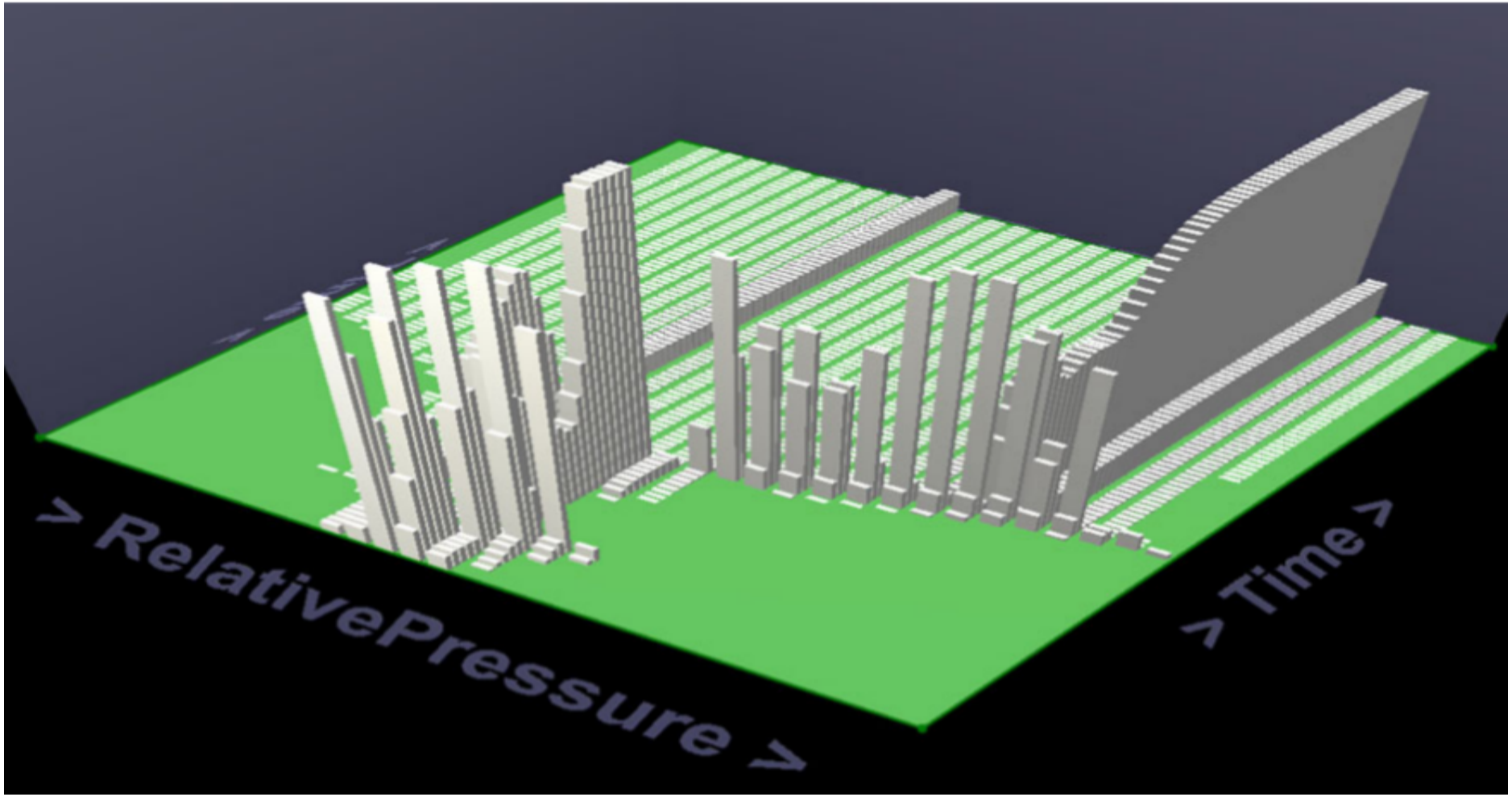
Interactive parallel bar charts



- x-axis = time; y-axis = hemodialysis sessions (patient); z-axis = blood pressure
  - [Chittaro et al. 03]

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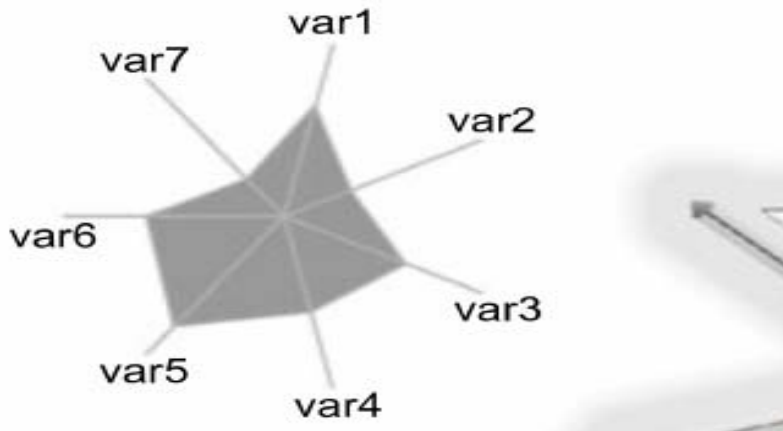
TimeHistogram 3D

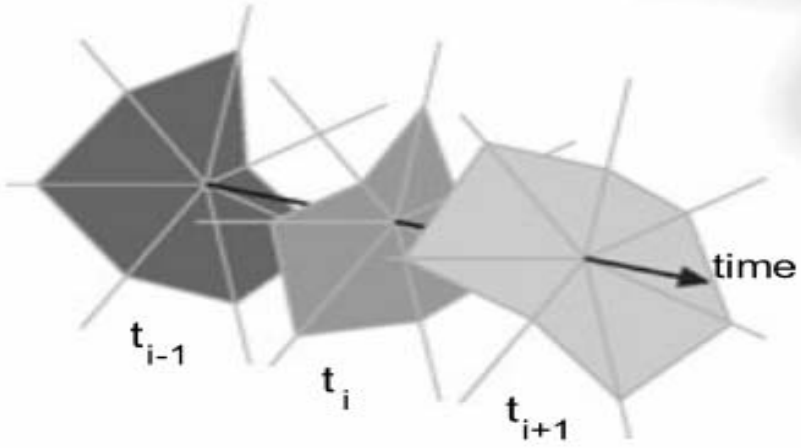


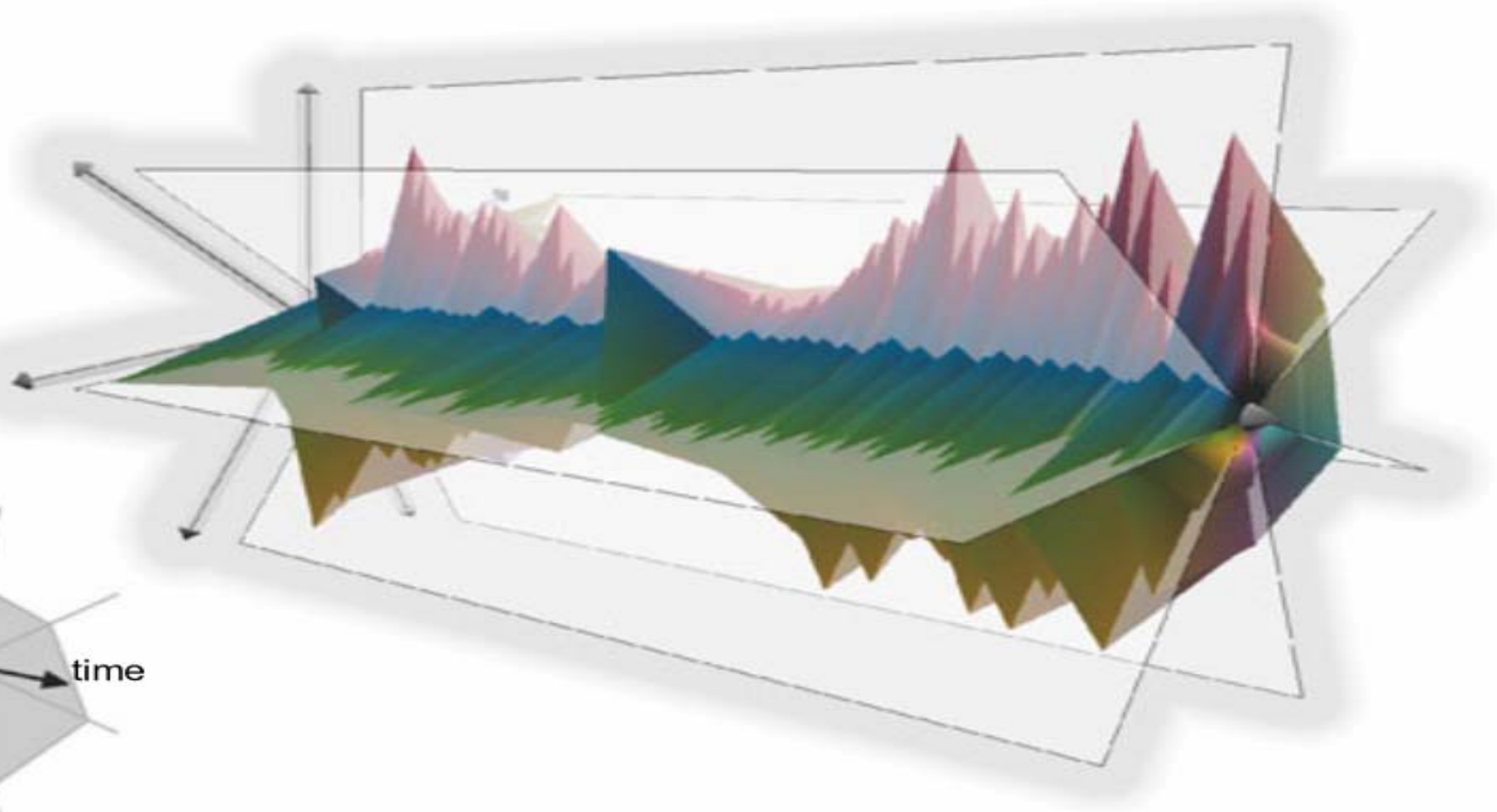
- x-axis = time; y-axis = pressure; z-axis = number of data items [Kosara et al 04]

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Kiviat tube



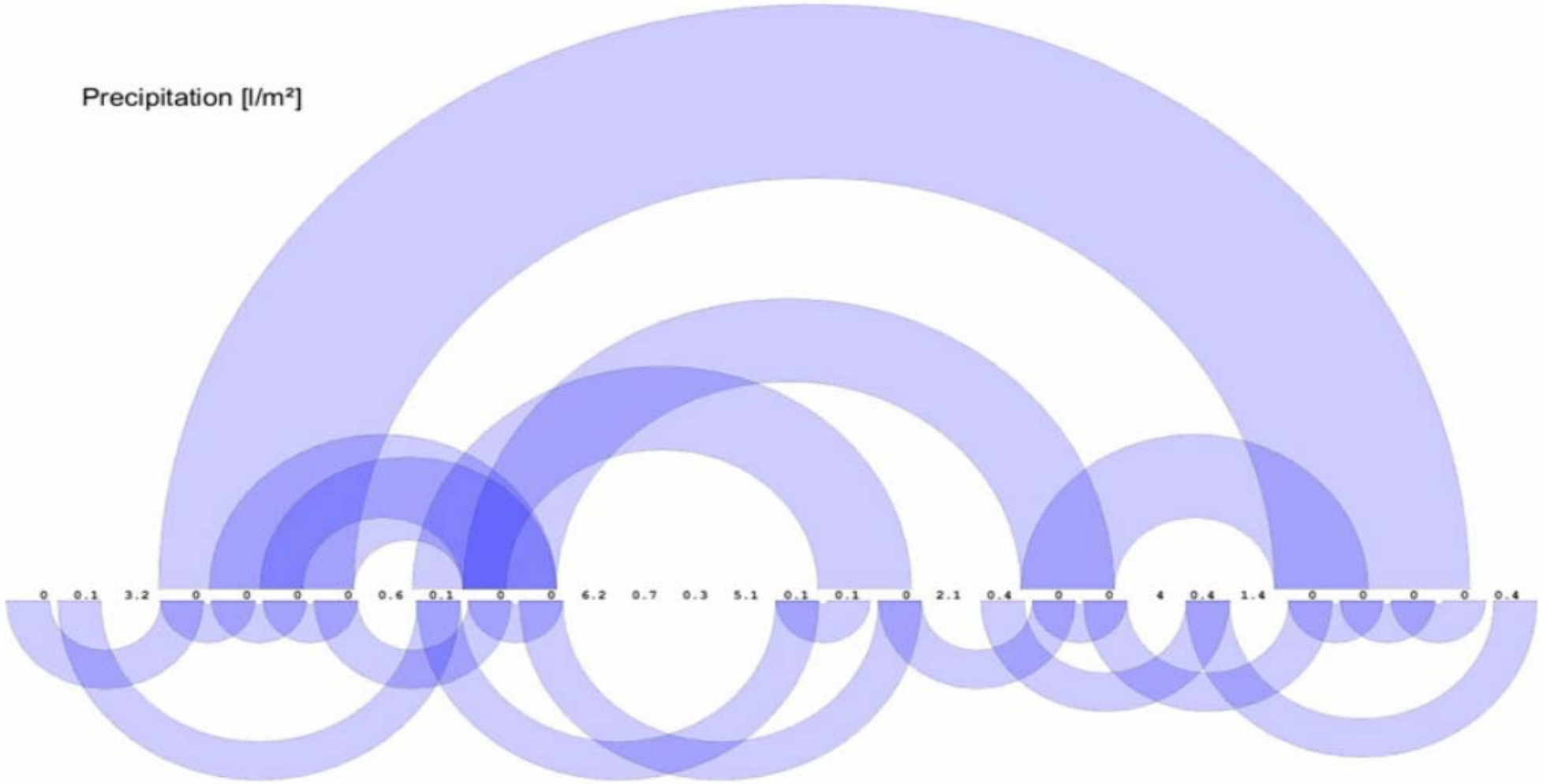




- Kiviat tube representing seven time-dependent variables
  - generated by the VisAxes3D tool [Tominski et al 05]

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Arc diagrams



- Preciptiation through time
  - [Wattenberg 02]

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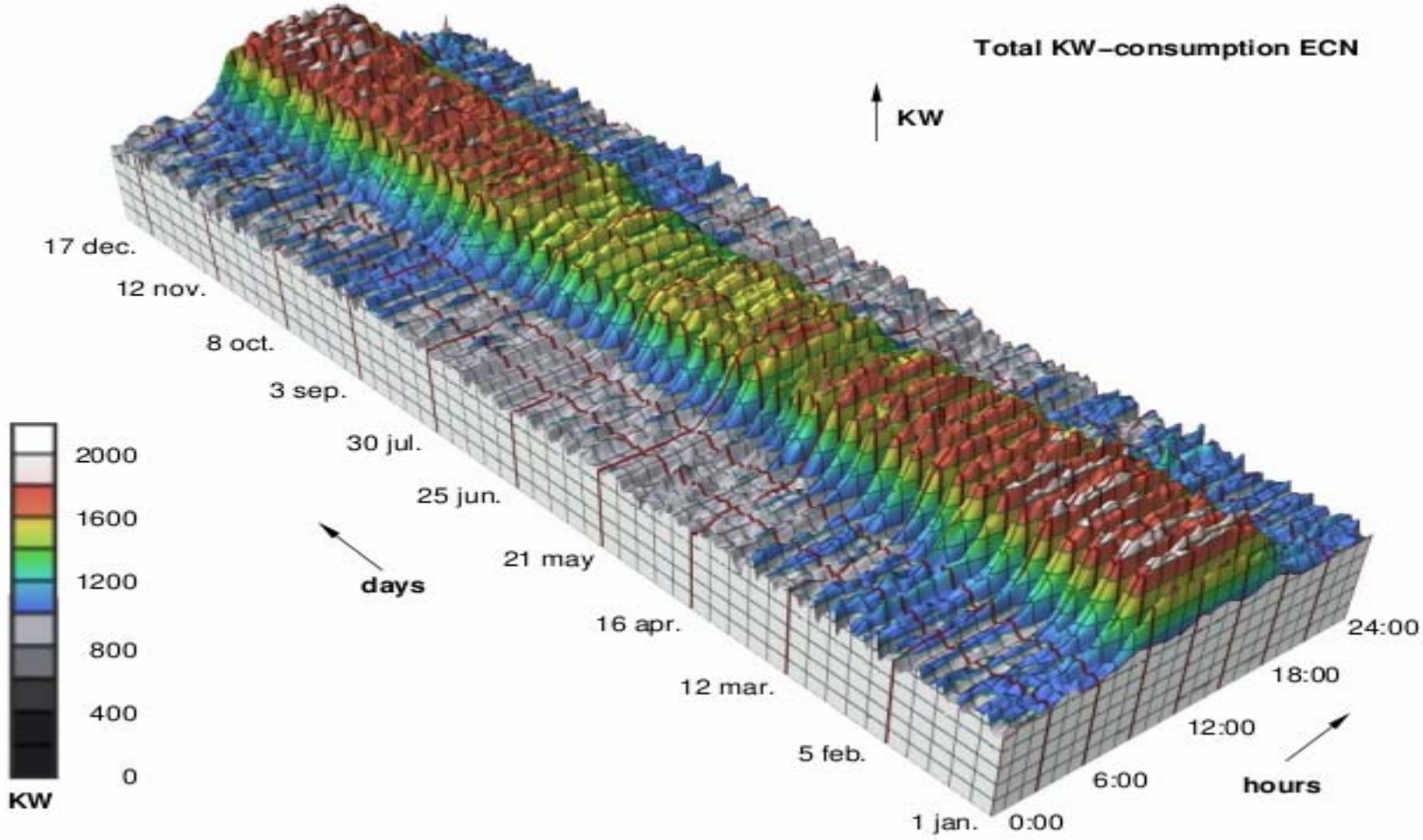
Discovering periodicity

- Example from [van Wijk & van Selow, ‘99]
- Data (two different timeseries)
  - power plant electricity usage over one year (1997)
  - workers punch in and punch out of a factory over one year (1997)
- Goal
  - understand the presence of patterns

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Discovering periodicity

- Naive 3D representation
- Visible features
  - seasonal trends
  - day-patterns
- Problems
  - occlusion
  - perspective distortion
  - variation over the week?
  - day-patterns for Sat and Sun?
  - details?



Picture from [van Wijk & van Selow, '99]

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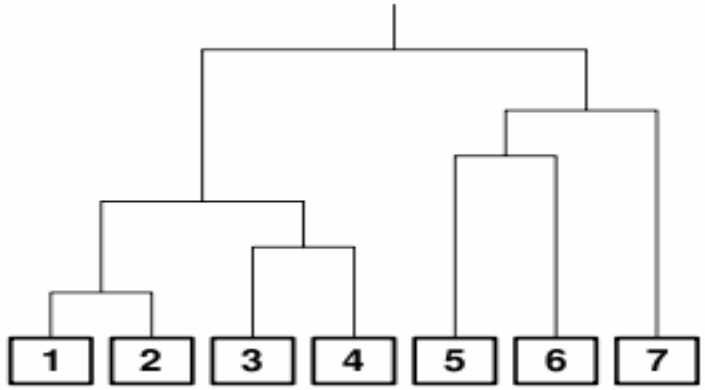
Alternative solution

- Cluster analysis combined with a calendar based visualization
  - cluster analysis
    - merge similar day patterns into clusters
  - calendar based visualization
    - visualize a representative for each cluster and the set of days it corresponds to

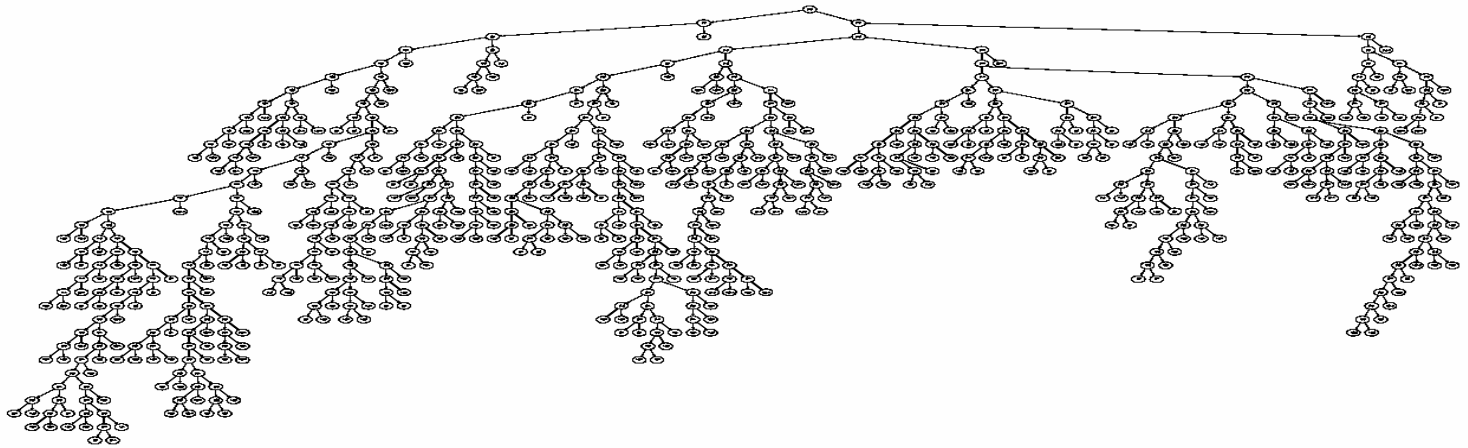
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Cluster analysis

- Merging process
  - find the two most similar days
  - merge them into one new cluster
  - keep repeating until some ending condition is reached



- The merging process produces a tree
  - called dendogram
  - each leaf is a single day

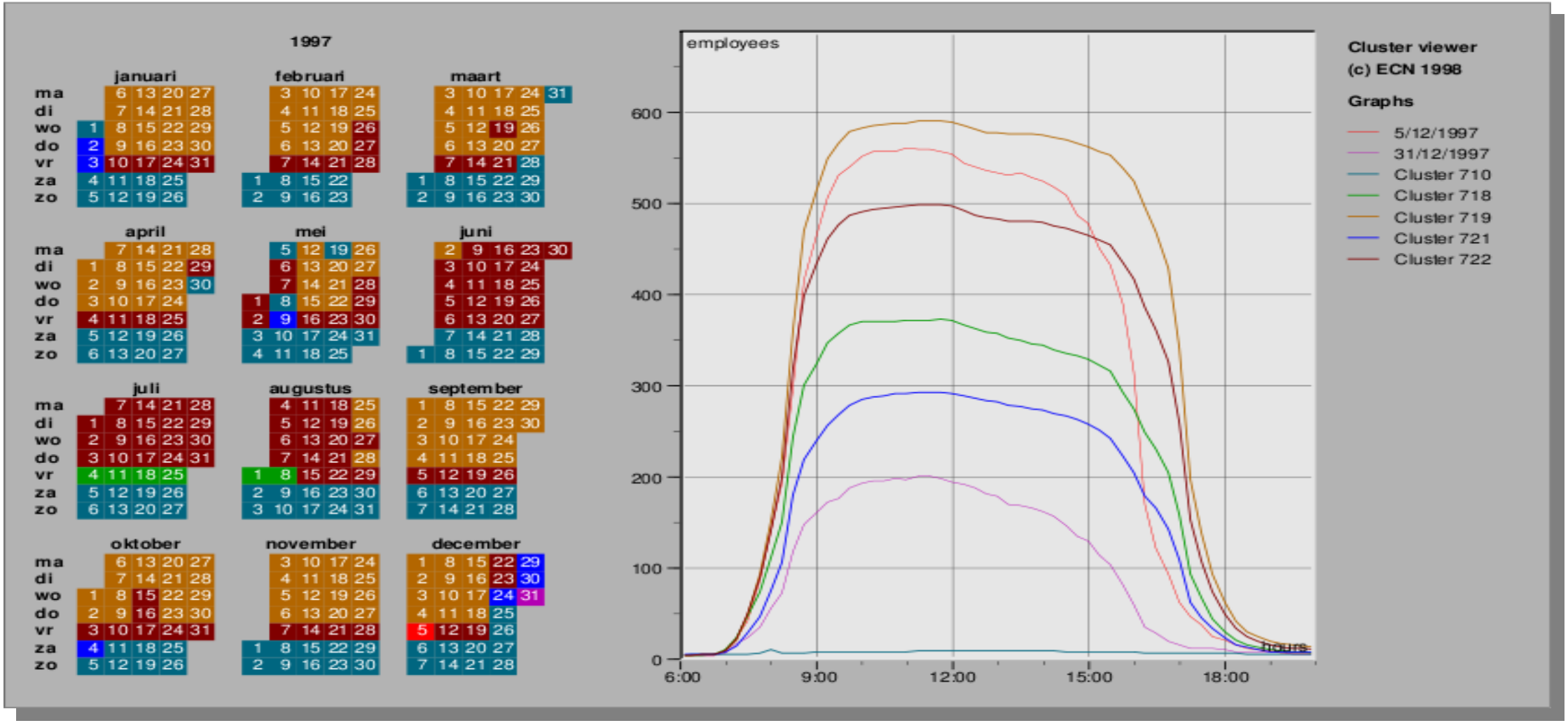


Pictures from [van Wijk & van Selow, '99]

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Calendar visualization

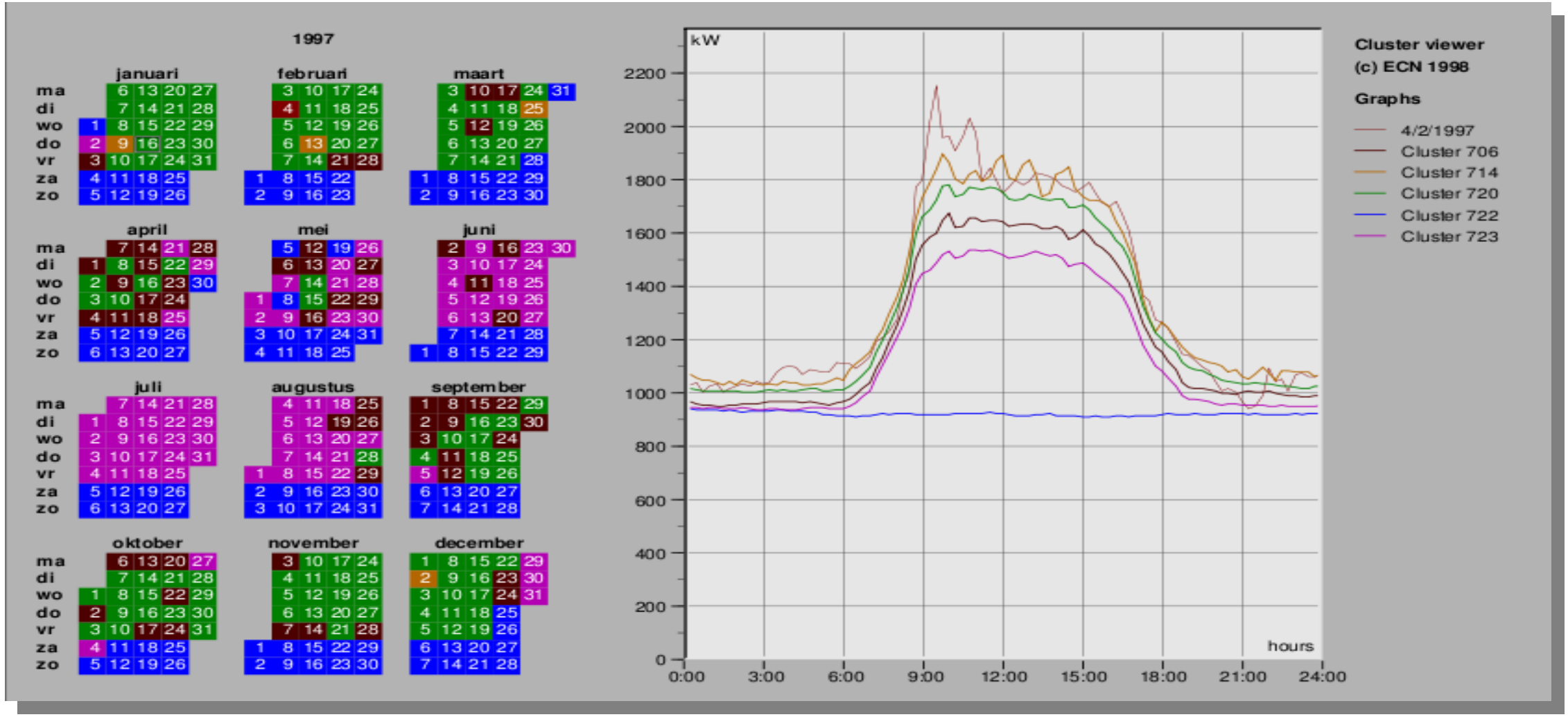
- Calendar view of the number of employees



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Calendar visualization

- Calendar view of power demands



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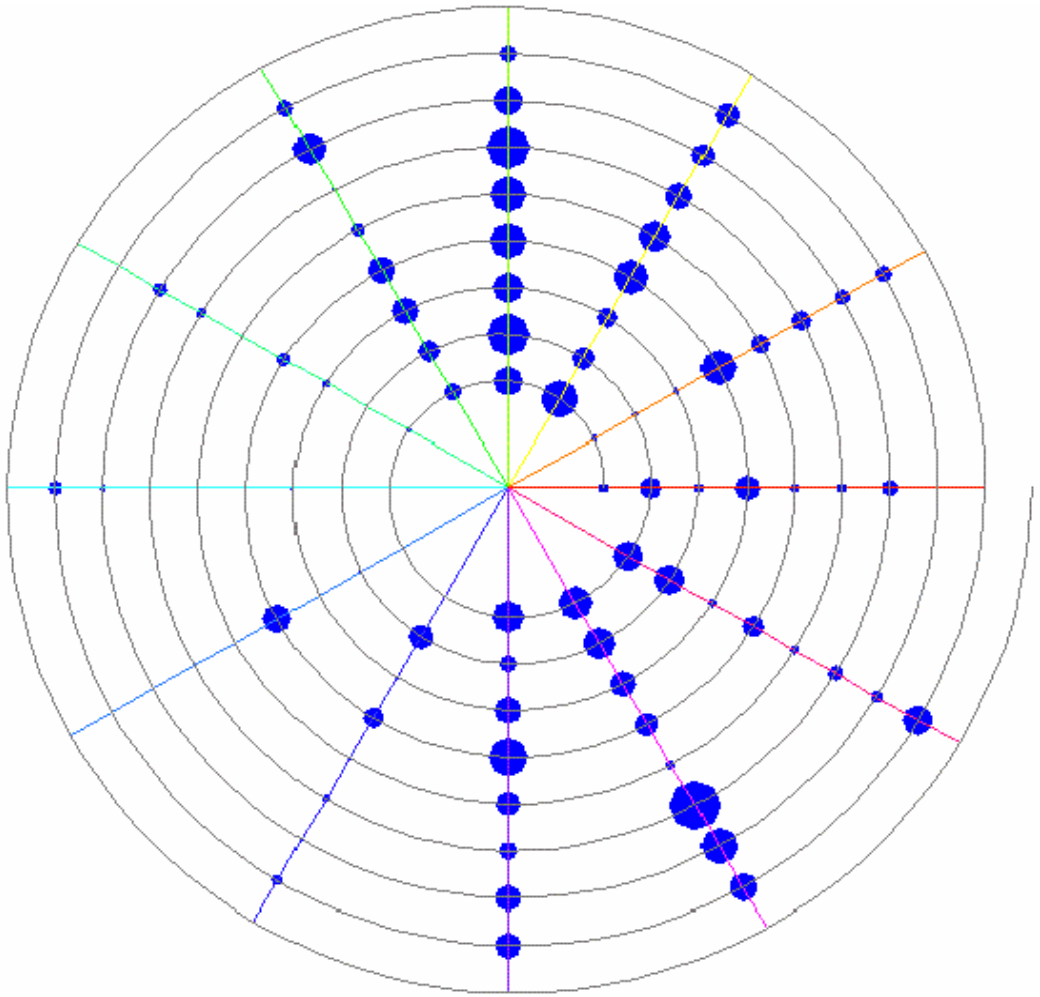
Interaction

- Selection
  - clicking on a day/month/year on the calendar toggles the corresponding line graphs in the display
  - clicking on a line graph highlights the corresponding day/days in the calendar
- Cluster expansion/contraction
  - select a line graph (day or cluster)
  - use the “<“ and “>” keys to split it or replace it with parent
- Top down exploration
  - specify the number of clusters you want to see
- Bottom up exploration
  - start from an empty line graph and select a day

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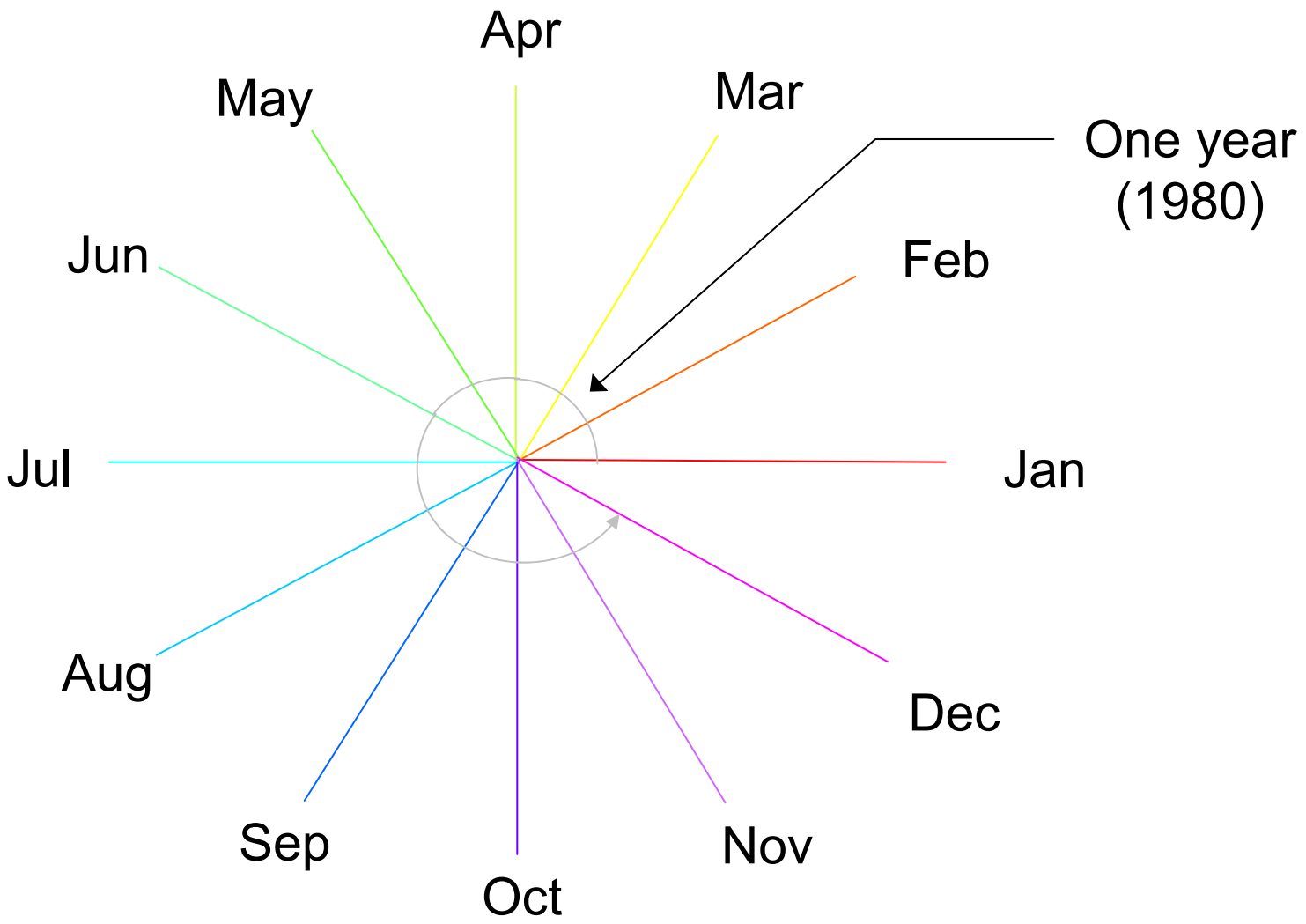
Discovering periodicity

- Year periodicity?
  - consumption of Baphia Capparidifolia by chimpanzees in the Gombe preserve in Tanzania for the years 1980 through 1988
    - From [Carlis & Konstan, ‘98]



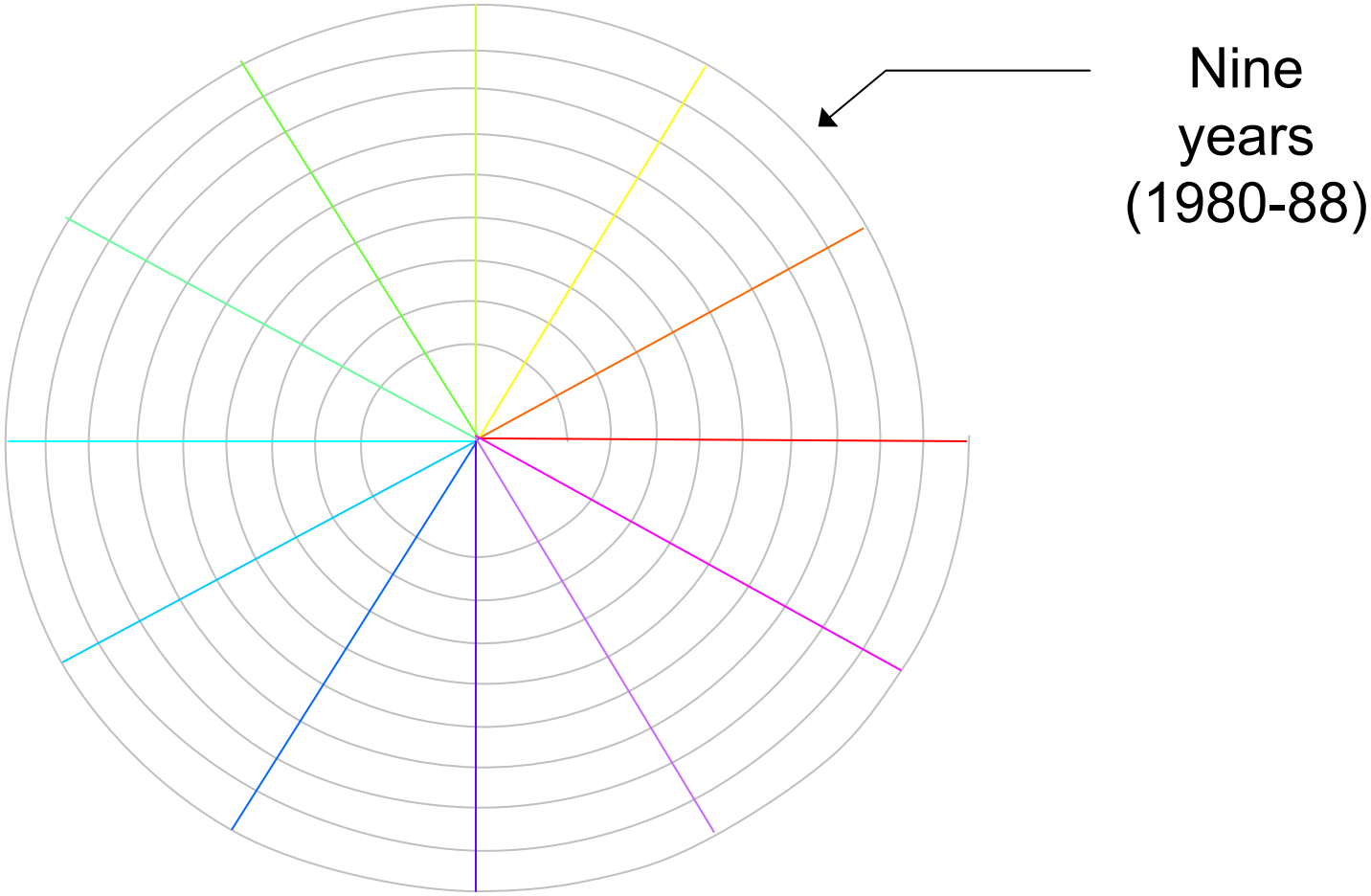
copyright ©2026 m.patrignani

Spiral representation



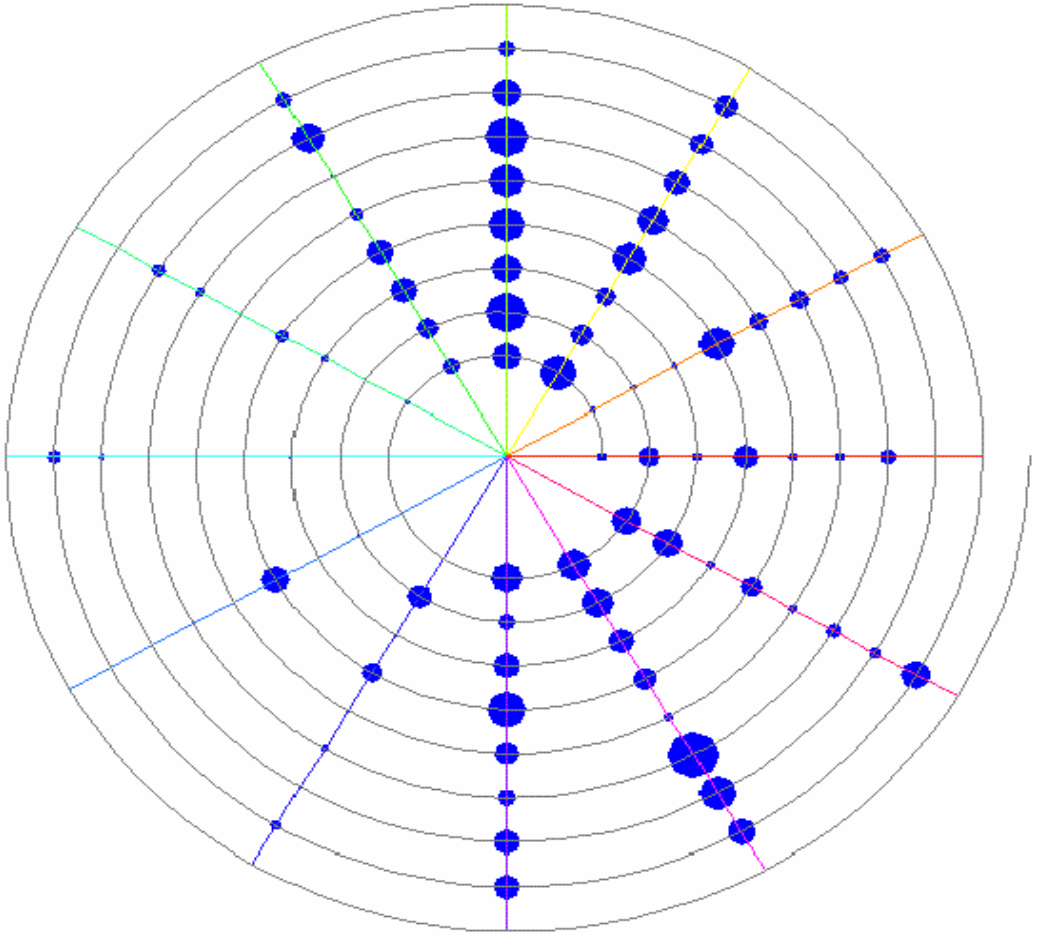
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Spiral representation



 copyright ©2026 m.patrignani

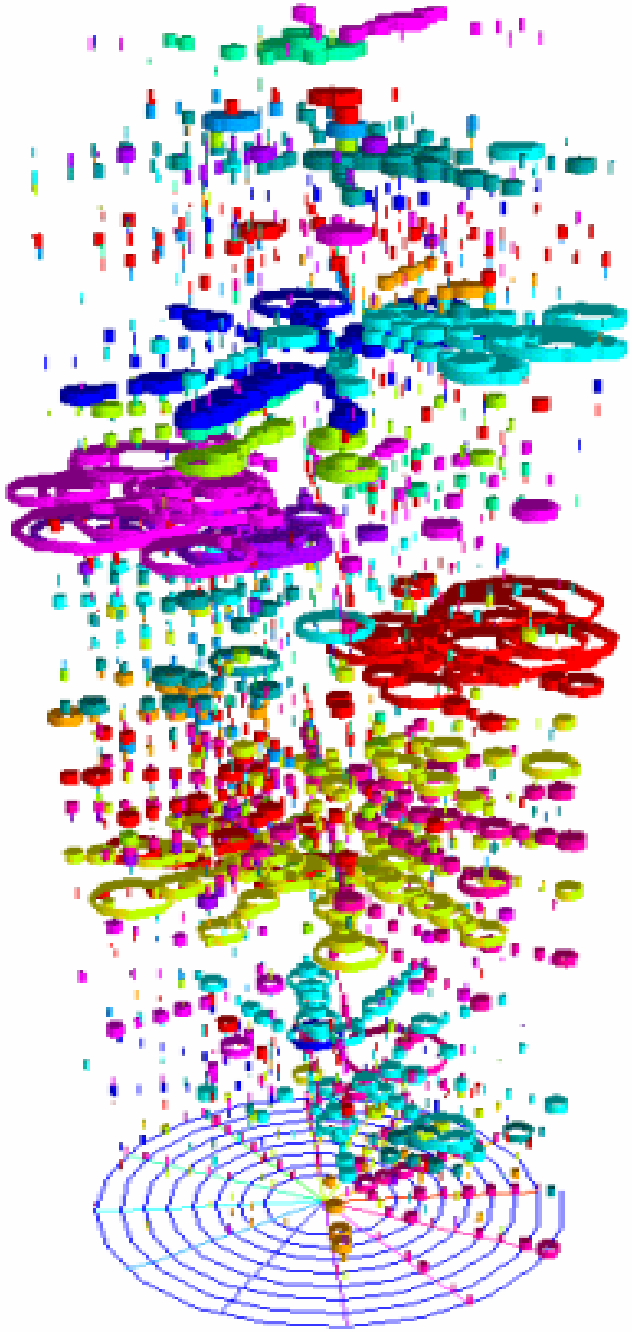
Spiral representation



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Periodicity

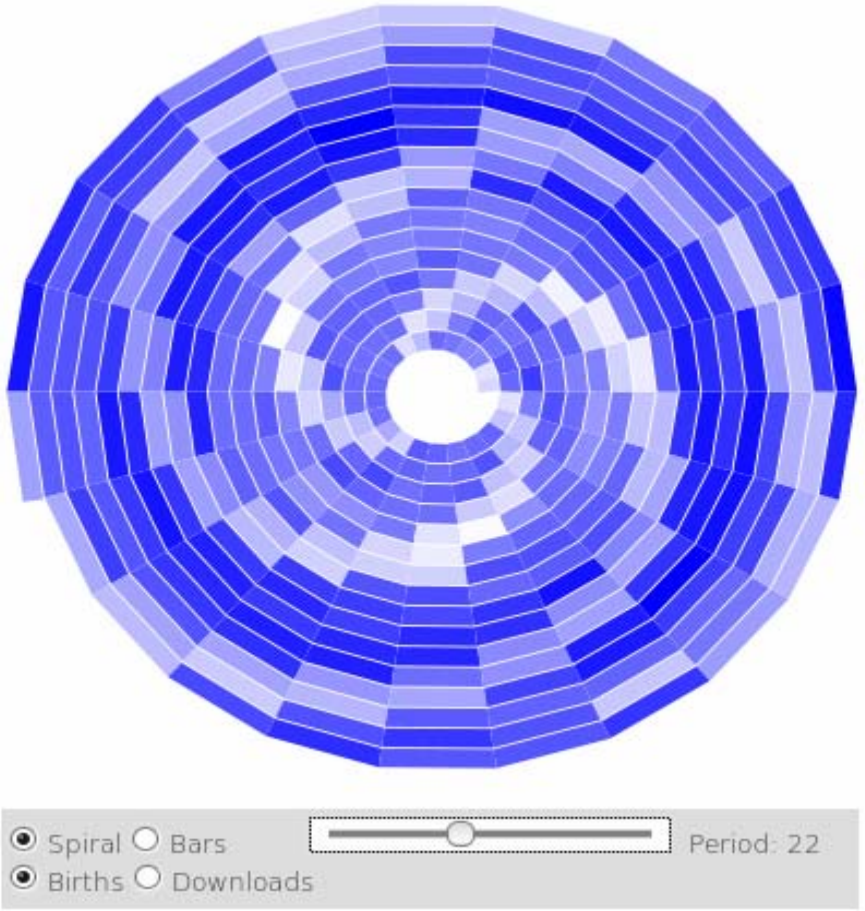
- Year periodicity
- Different foods represented by different colors and drawn at different heights
- Can you still see serial and periodic attributes?
- As with all 3-D, requires navigation
  - from [Carlis & Konstan, ‘98]

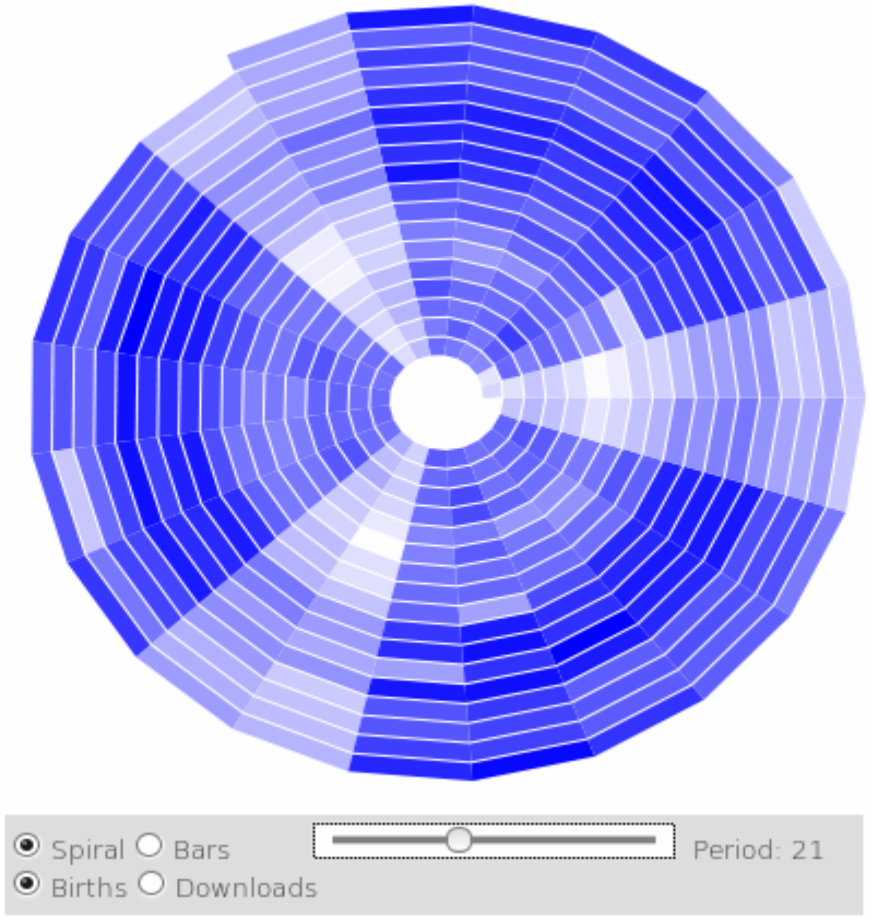


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Spirals for periodic data

- Using spirals to find periods



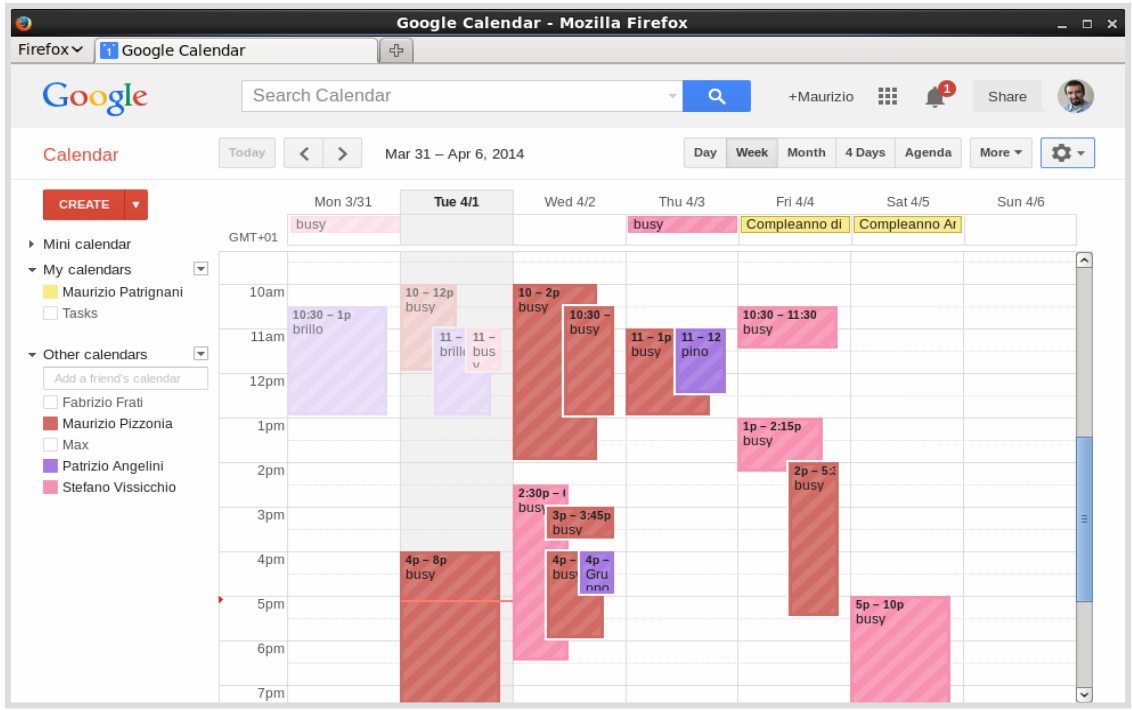


http://eagereyes.org/techniques/spirals

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Case 1: Calendar visualization

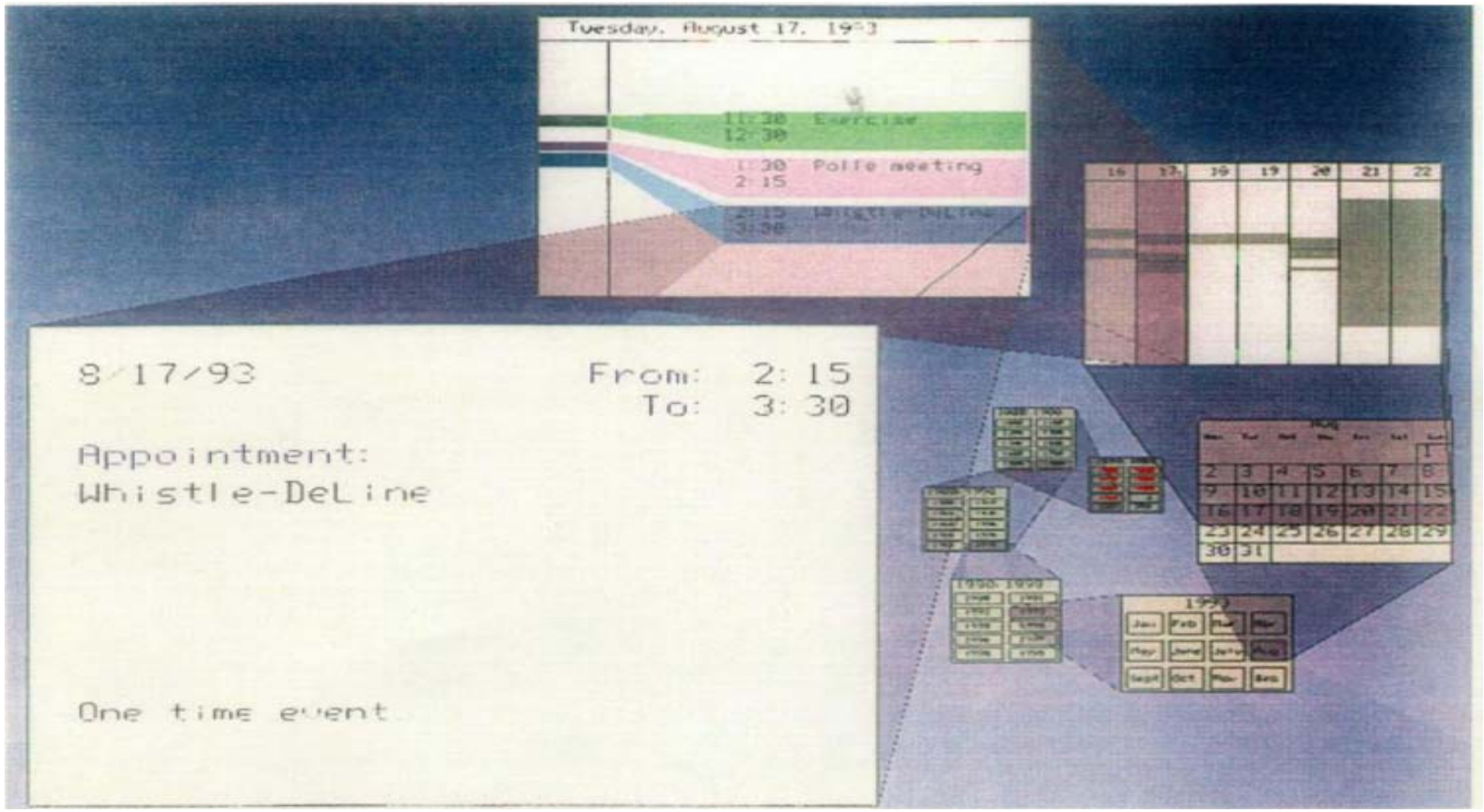
- Visualization of personal calendars
- Common tasks
  - show both details and broader context
  - see commonly available times for group of people



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Spiral calendar

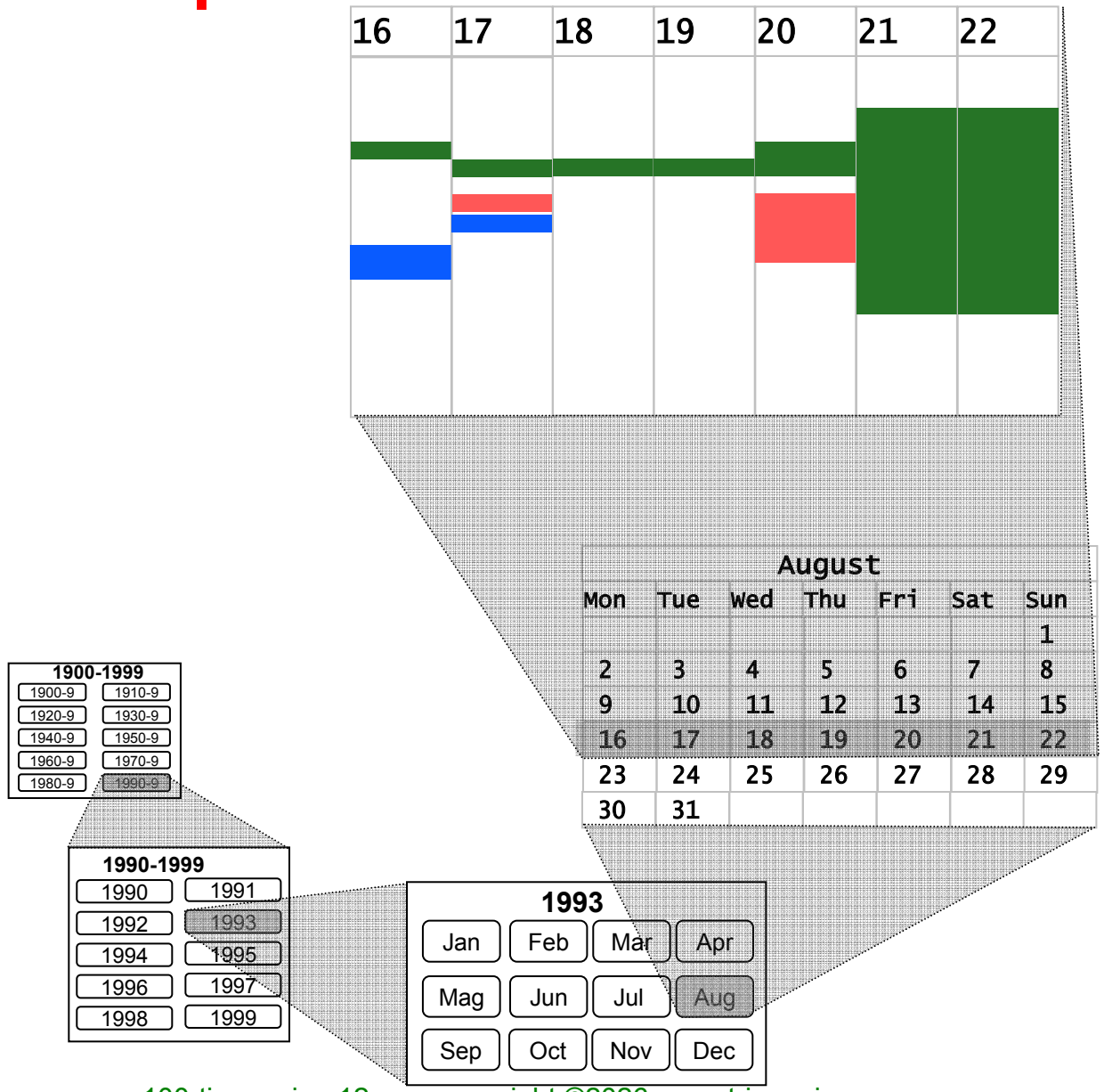
- Use 3D to open the hierarchy
  - event
  - day
  - week
  - month
  - year
  - decade
  - …



[Mackinlay *et al.*, 94]

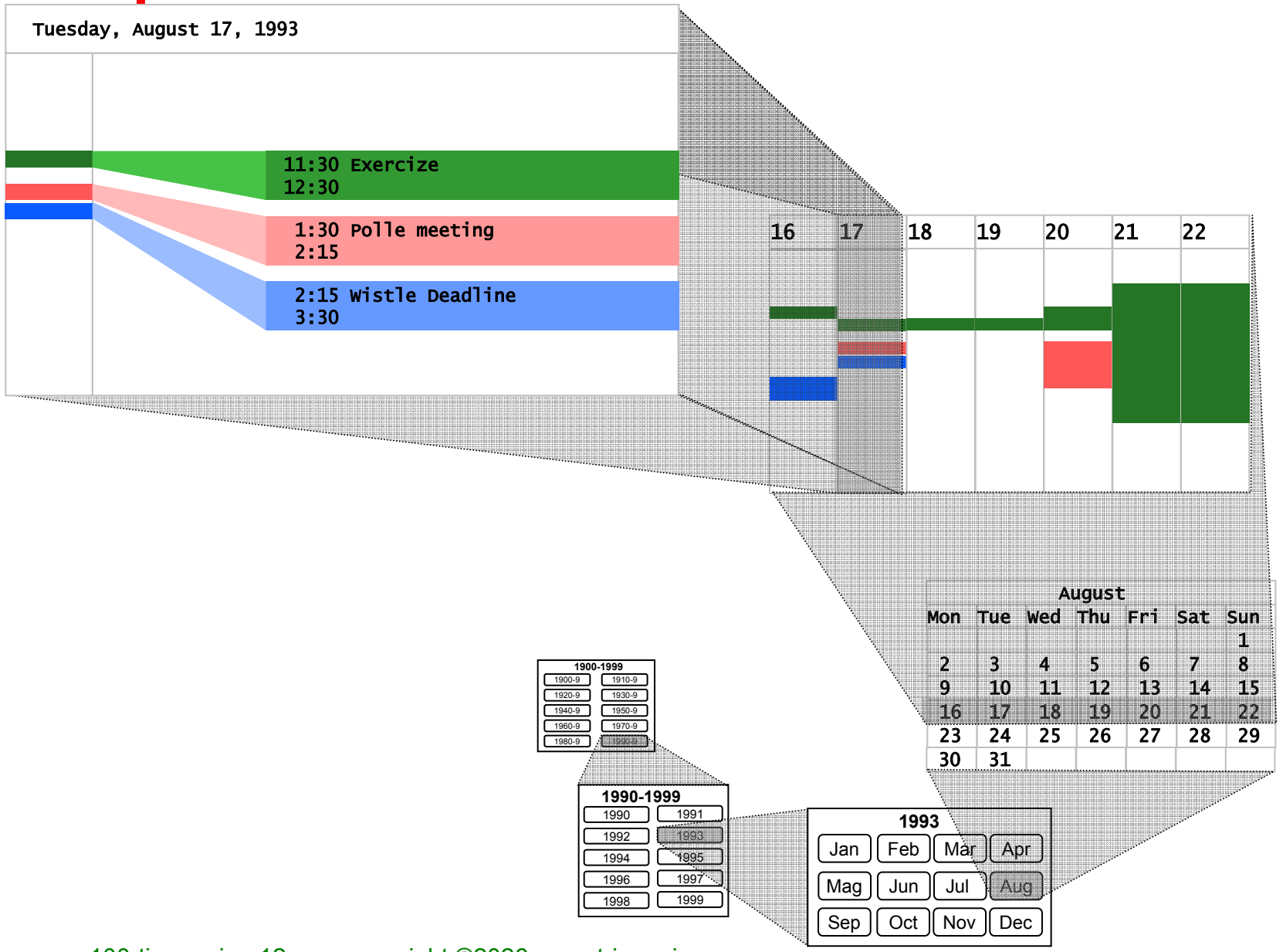
 copyright ©2026 m.patrignani

Spiral calendar



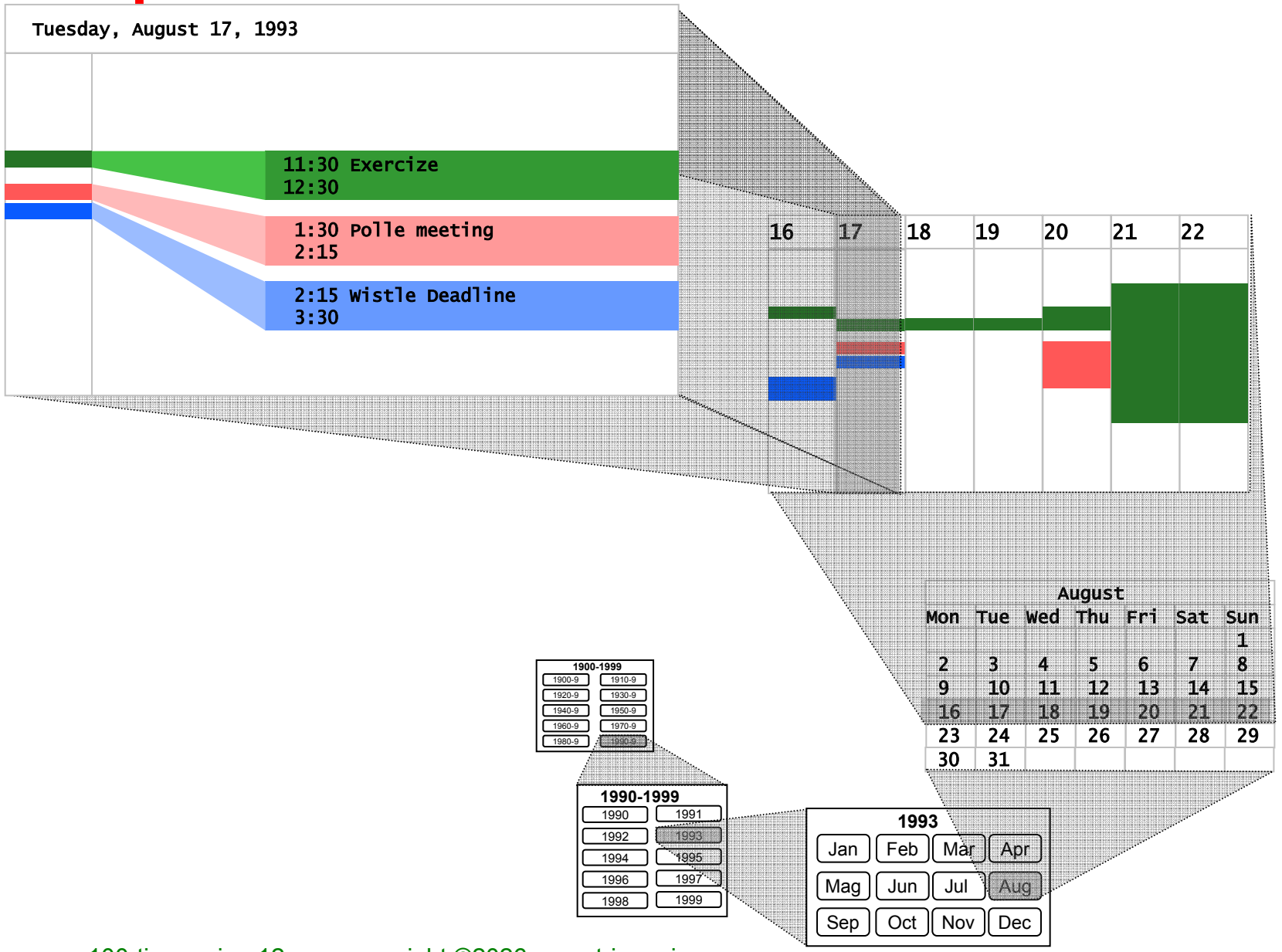
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Spiral calendar



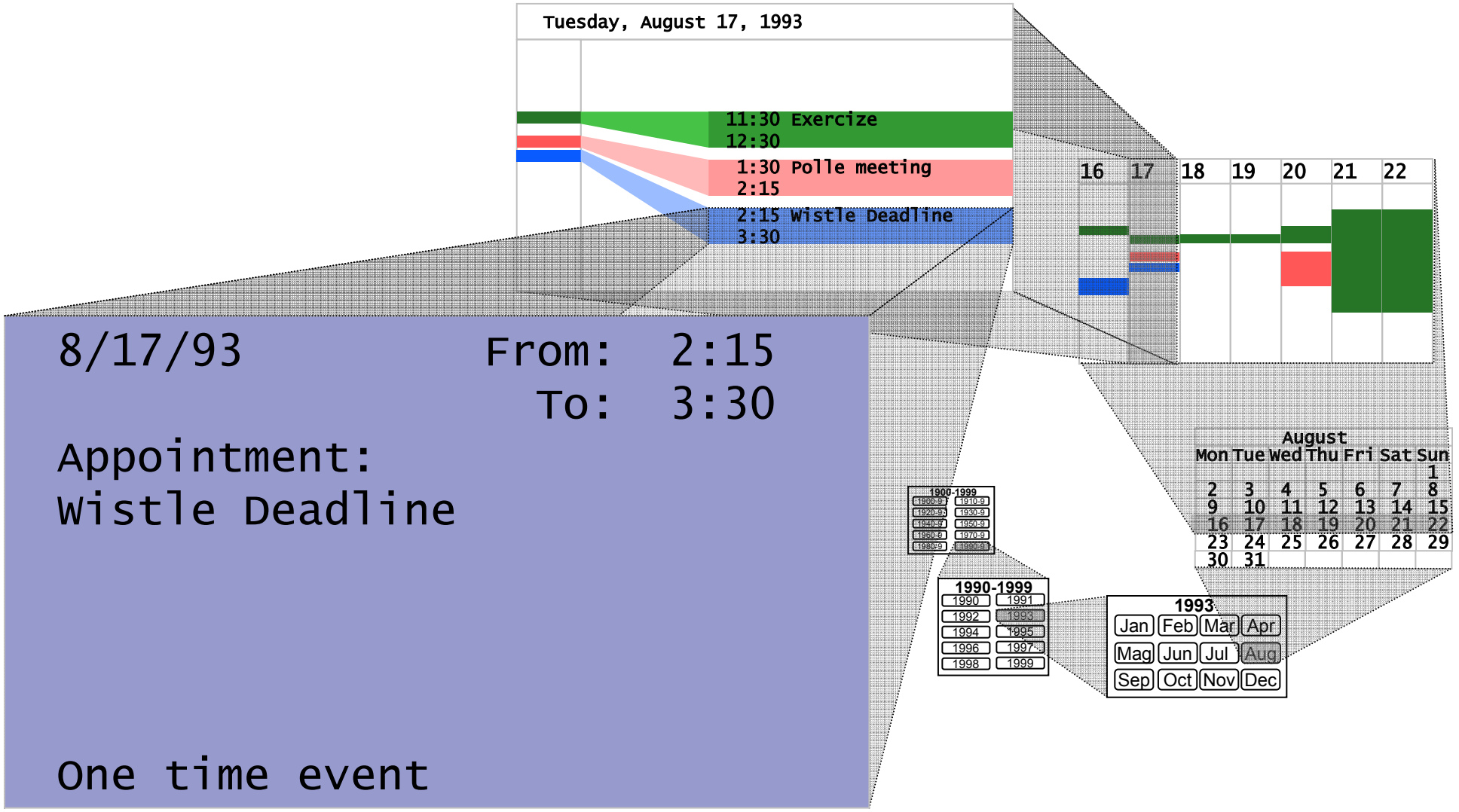
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Spiral calendar



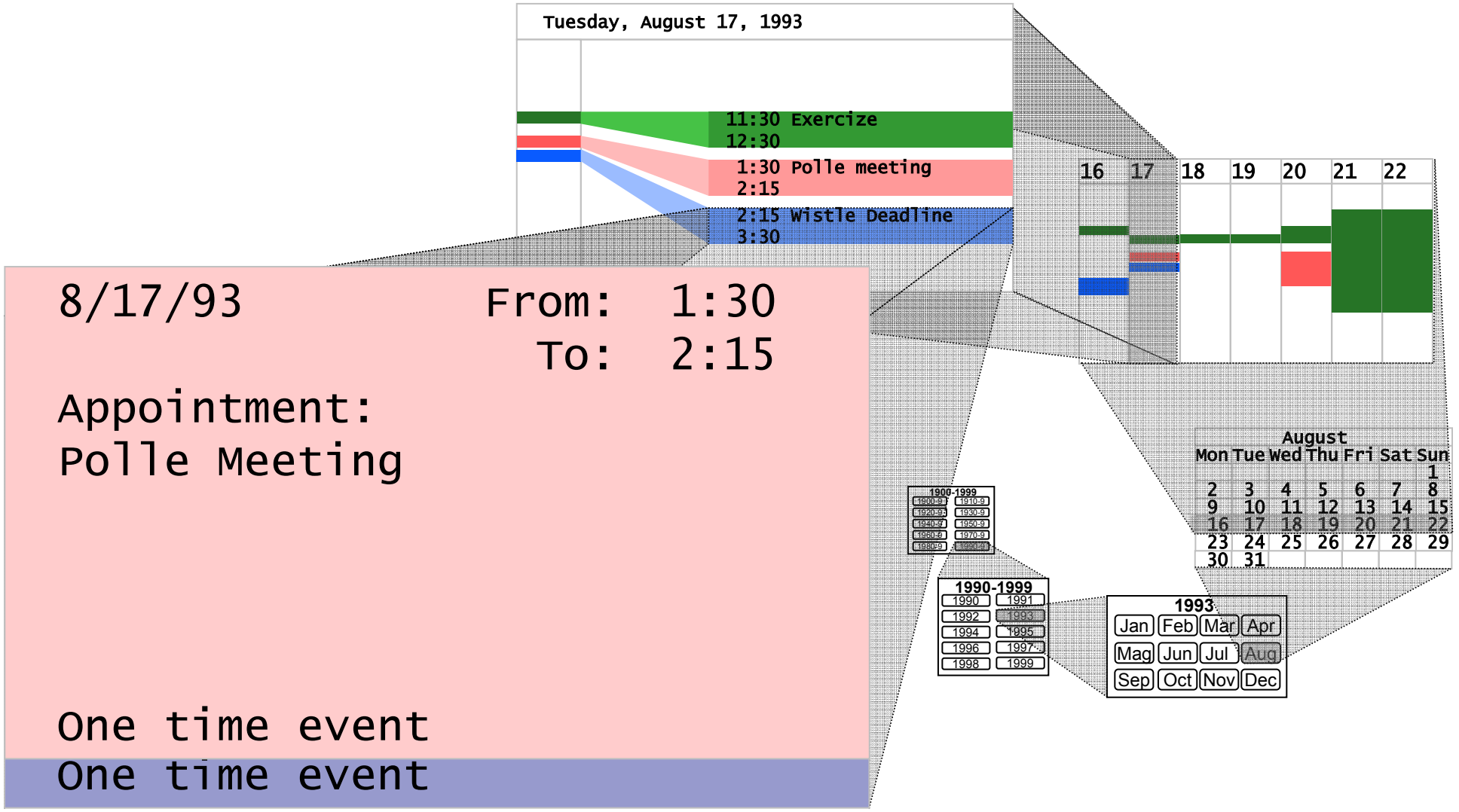
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Spiral calendar



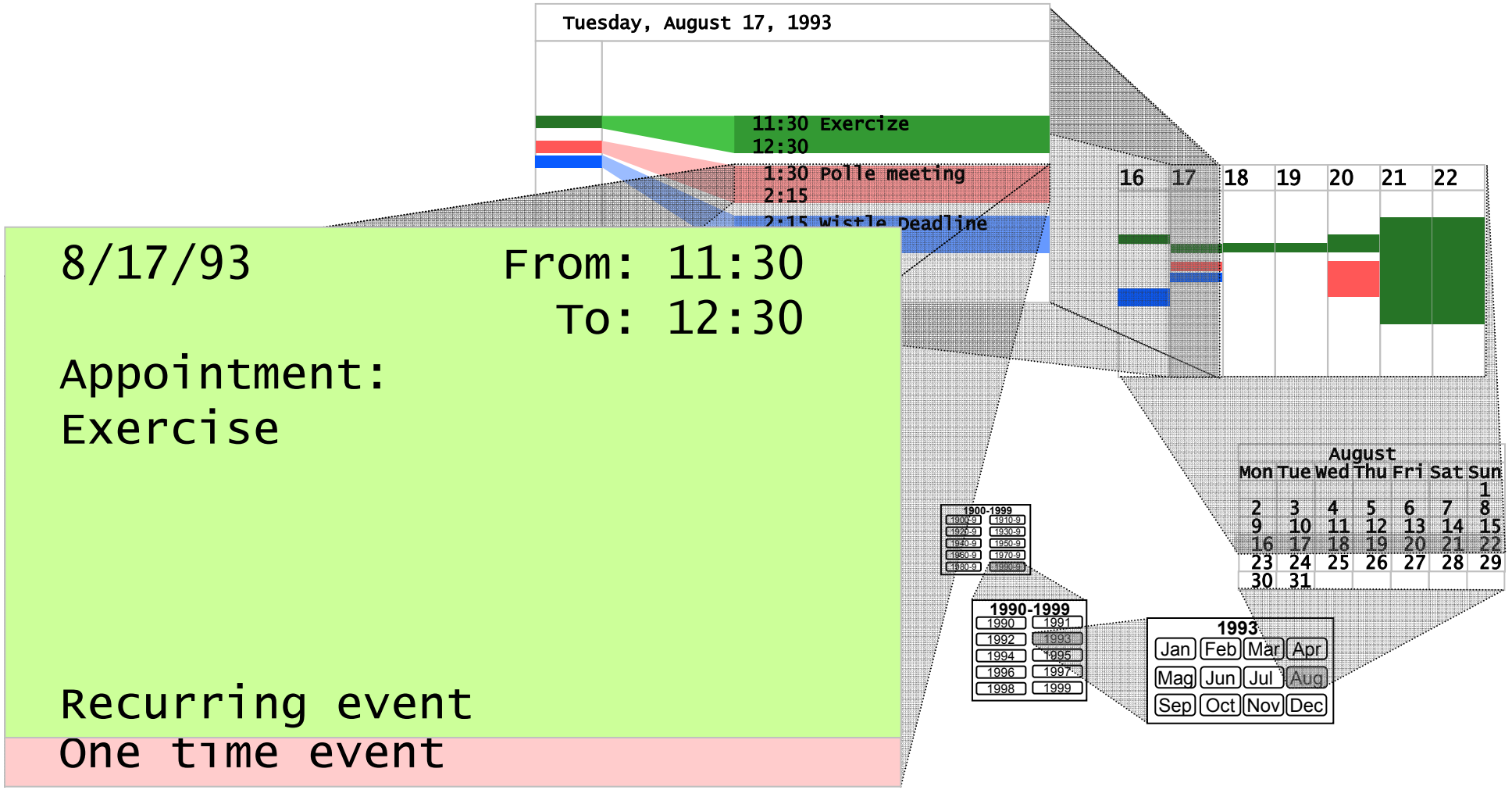
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Spiral calendar



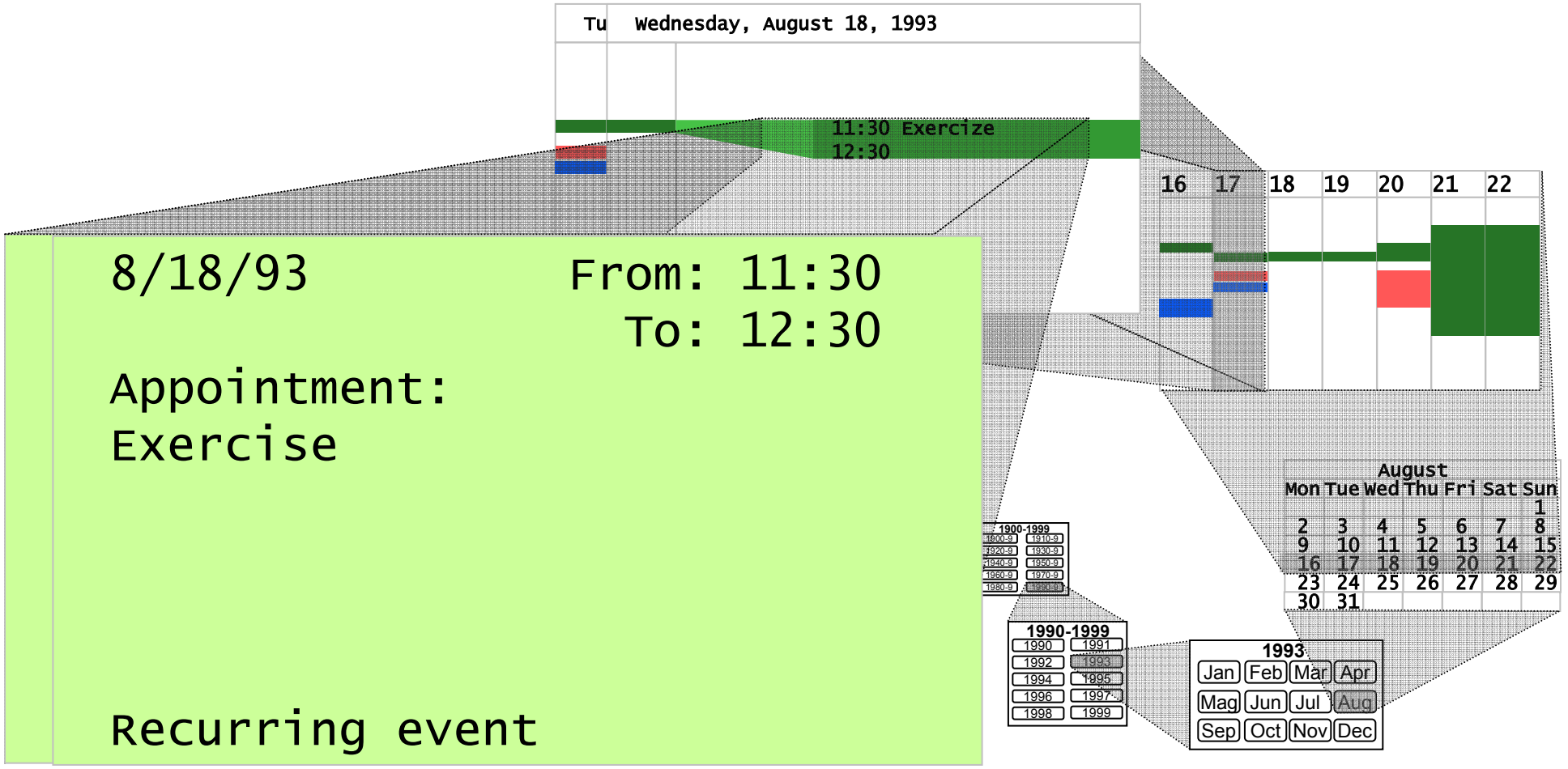
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Spiral calendar



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Spiral calendar



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Time lattice

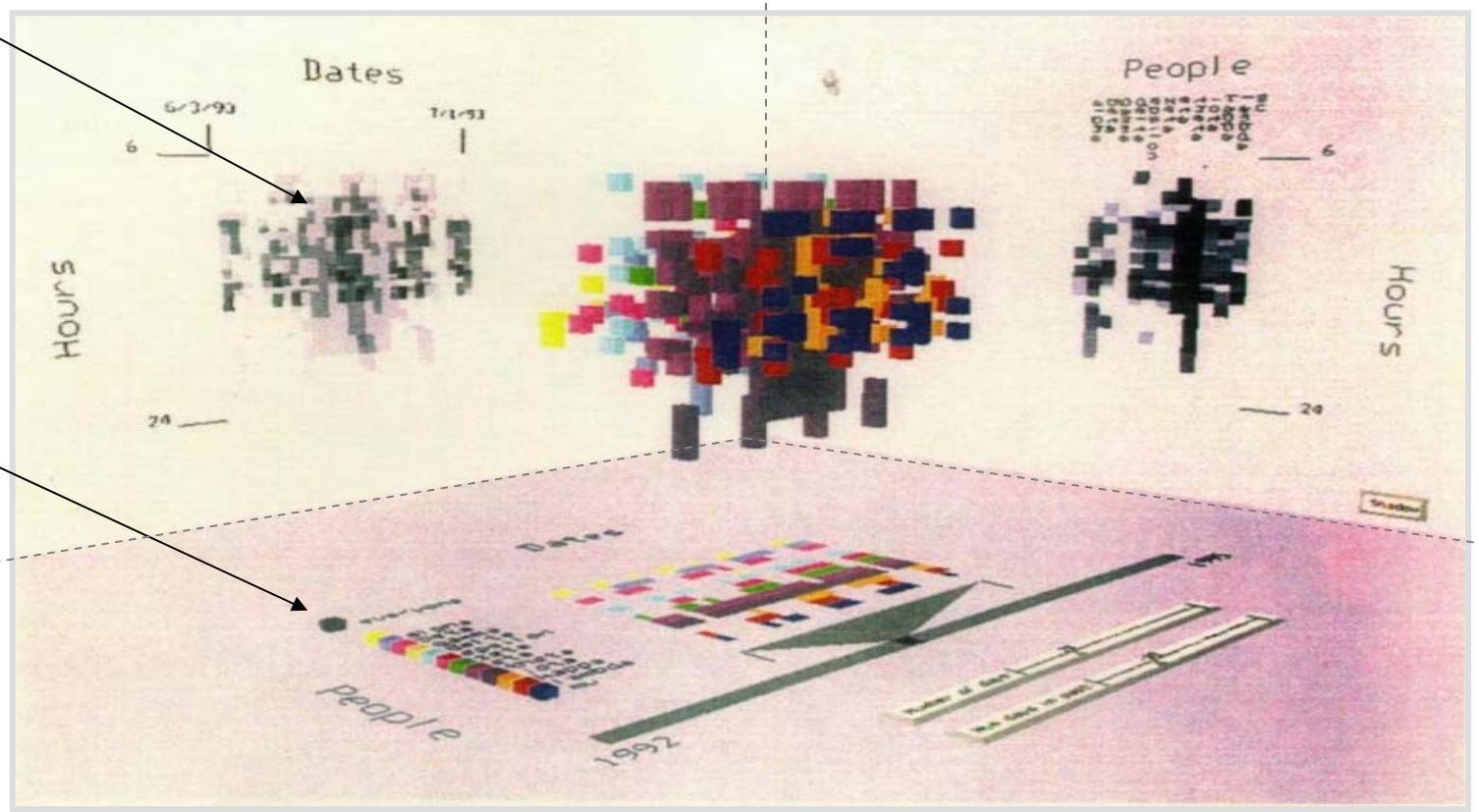
Empty spaces are available slots for meetings

Buttons to add/remove individual calendars

Hours

Dates

People

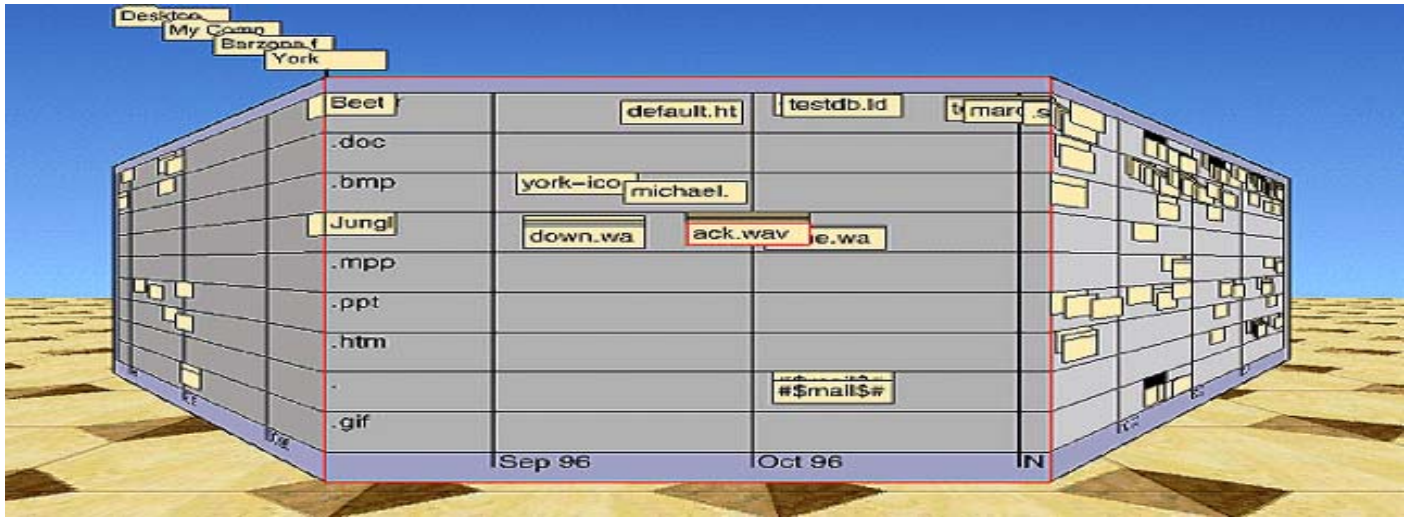


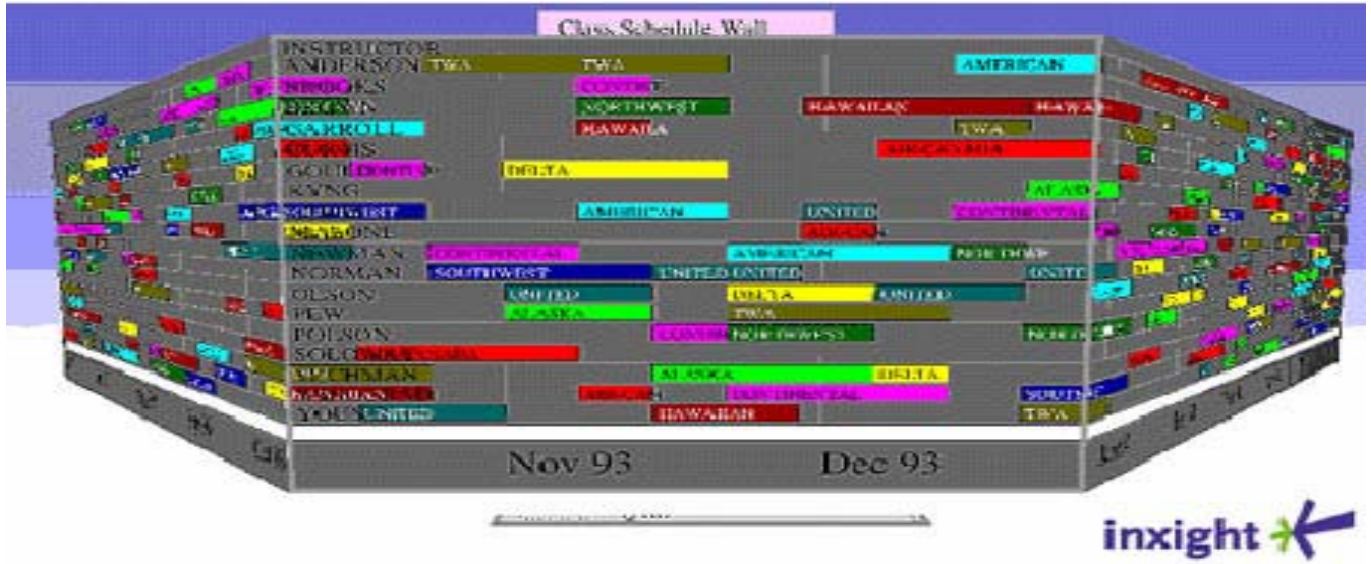
[Mackinlay *et al*., ‘94]

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Perspective wall

- The research version of the perspective wall
  - [Mackinlay *et al.*, 91]
- The version commercialized by Inxight Software
  - now acquired by Business Objects





inxight

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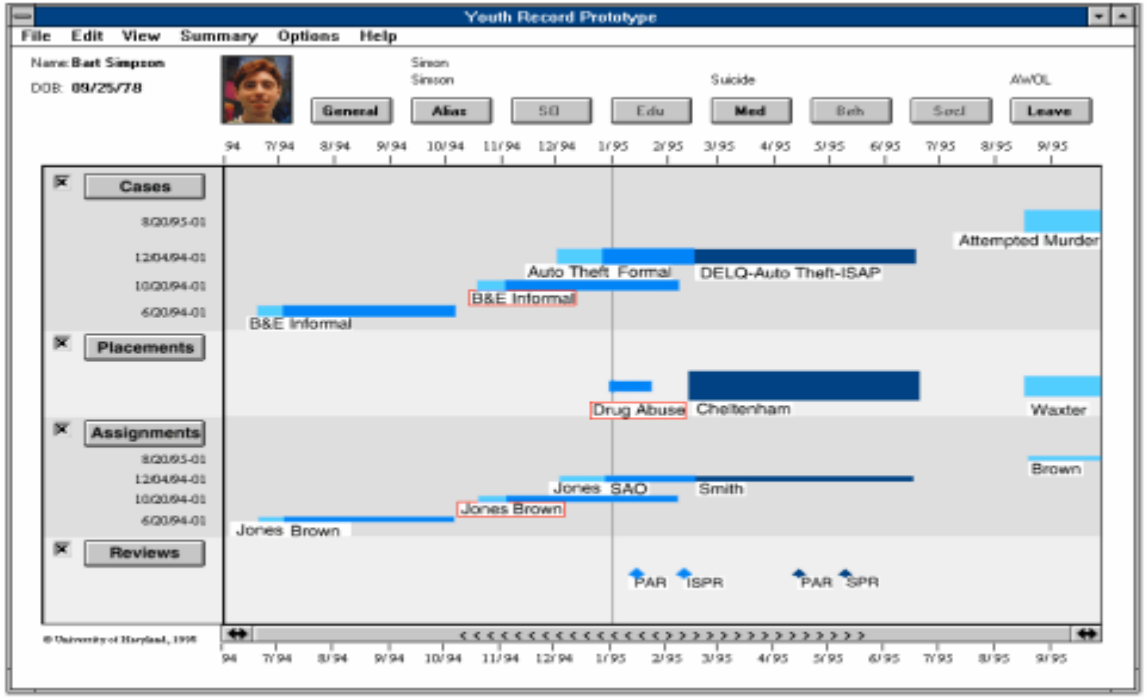
Case 2: Personal histories

- Consider a chronological series of events in someone's life
  - present an overview of the events
  - examples
    - medical history
    - educational background
    - criminal history
- Goals
  - put together complete story
  - gather information for decision-making
  - notice trends
  - gain an overview of the events

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LifeLines Project

- [Plaisant *et al.*, 96]
- Interaction
  - clicking on an event produce more information
  - clicking on some objects produces queries to a db

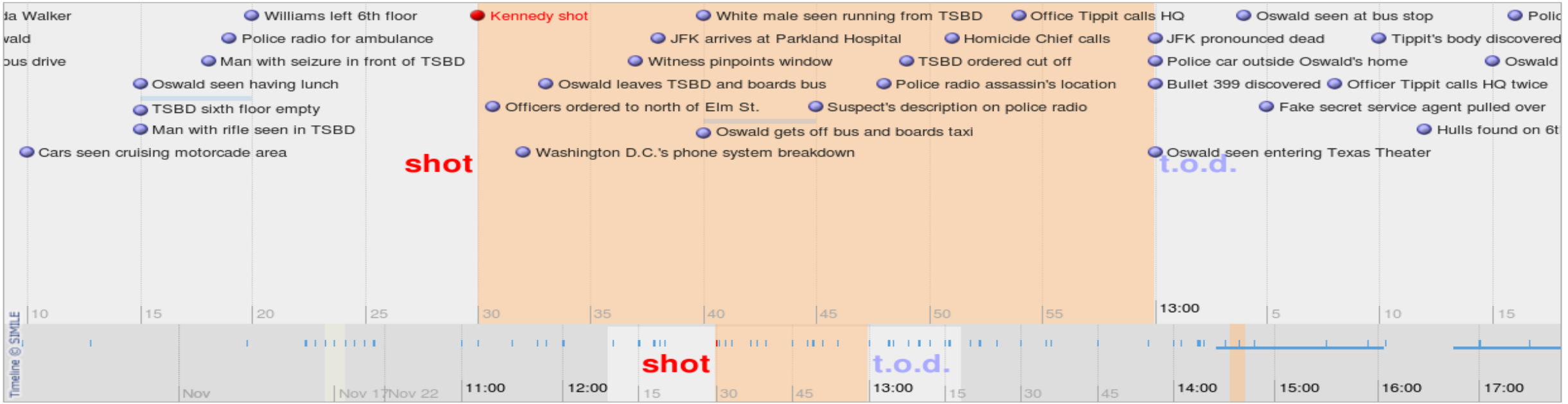


Video at http://youtu.be/aeRklur2Zc4

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SIMILE Timeline

- SIMILE Timeline
  - Open source widget



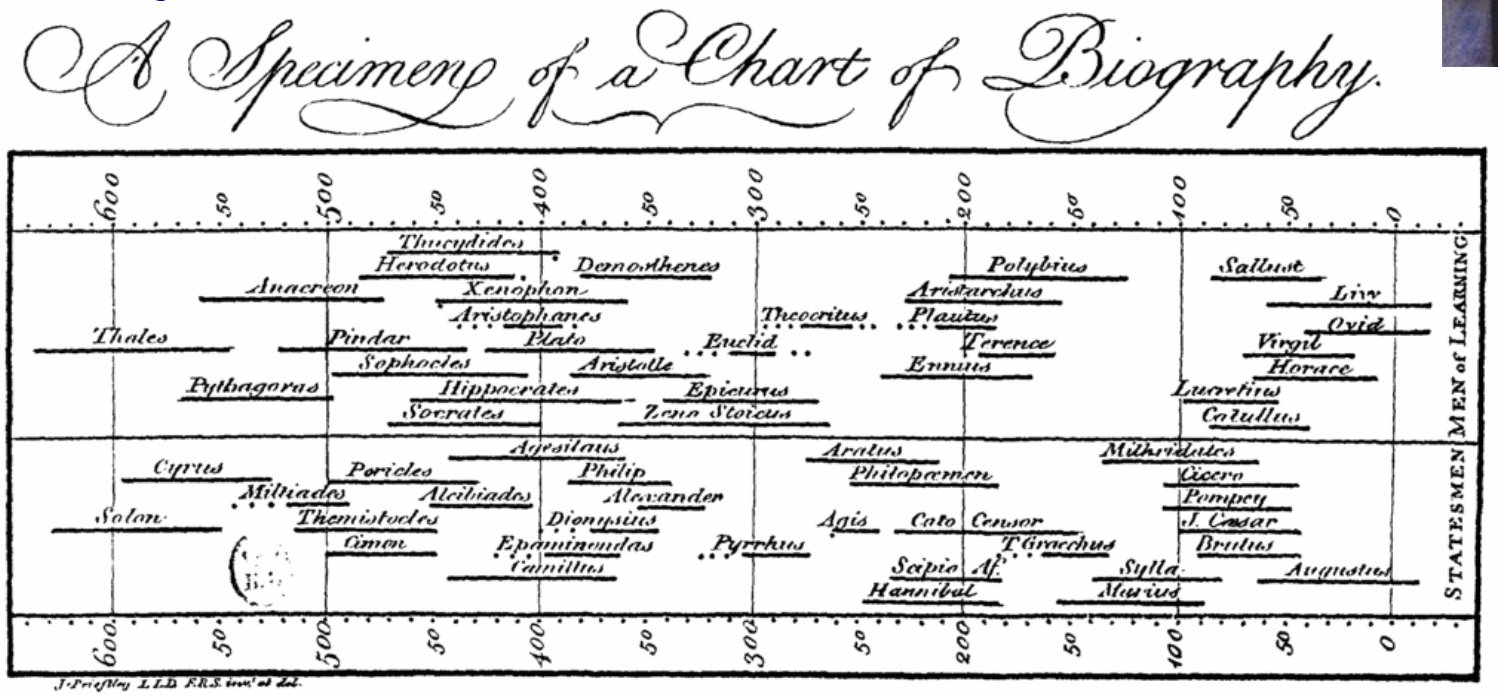
http://simile-widgets.org/timeline/

 copyright ©2026 m.patrignani

Who invented timeline charts

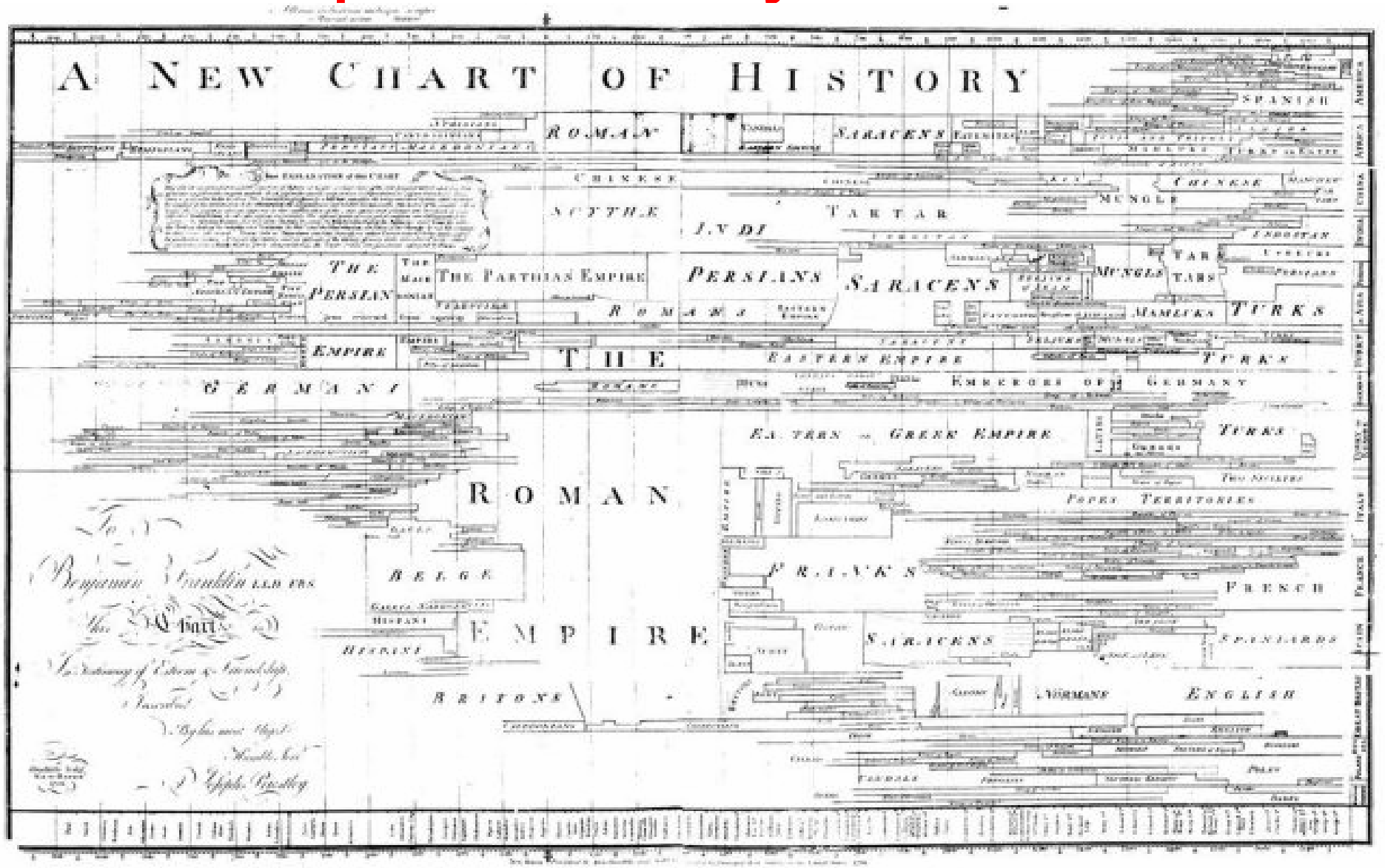
- Joseph Priestley (1733-1804) is credited with the first timeline charts in 1765
  - each bar represents the life span of a person
- Priestley's work was an inspiration for William Playfair's





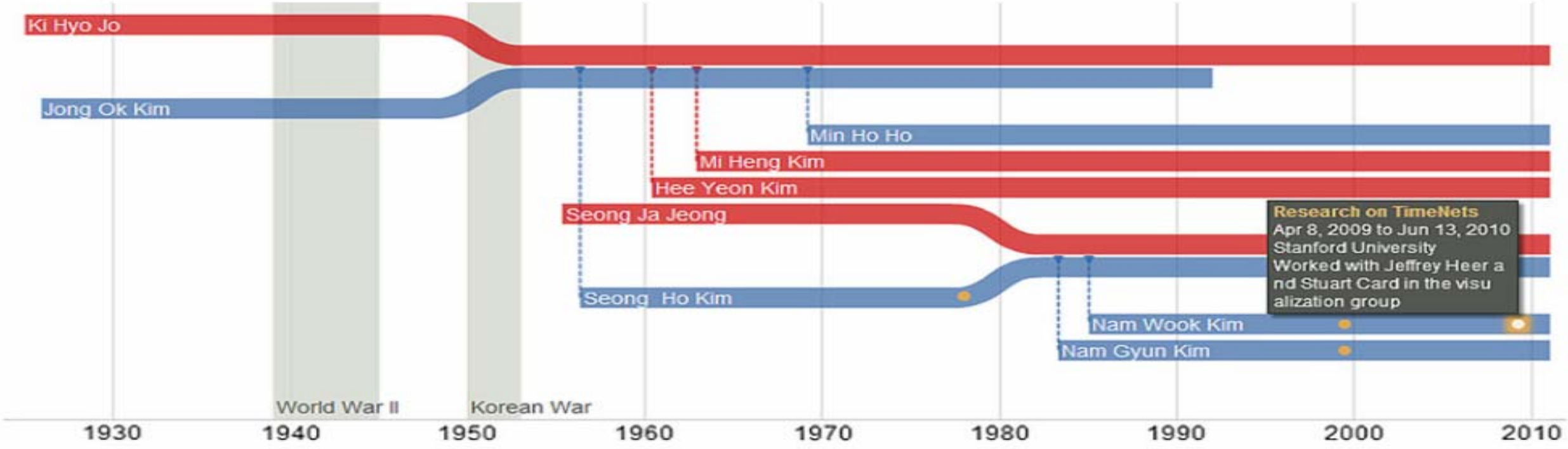
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Joseph Priestley's timelines



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TimeNets



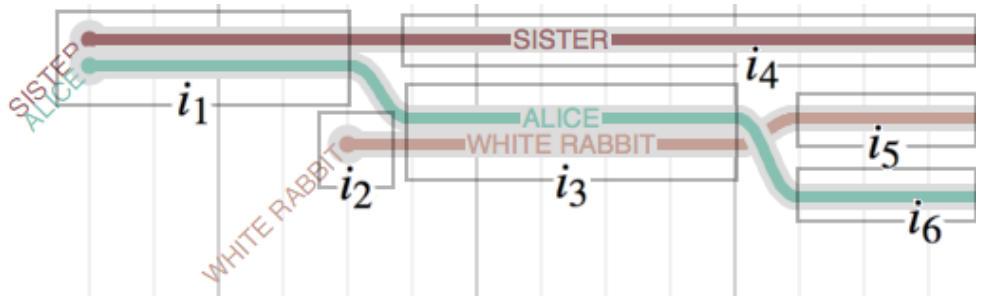
- Visualizes temporal and structural aspects of genealogical data
  - [Kim et al 10]

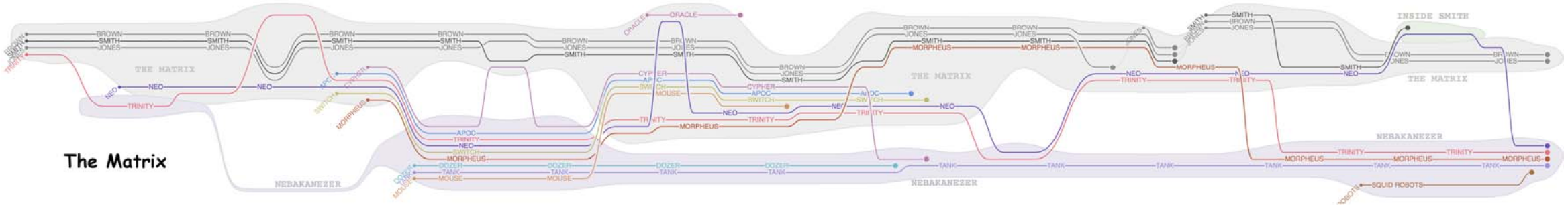
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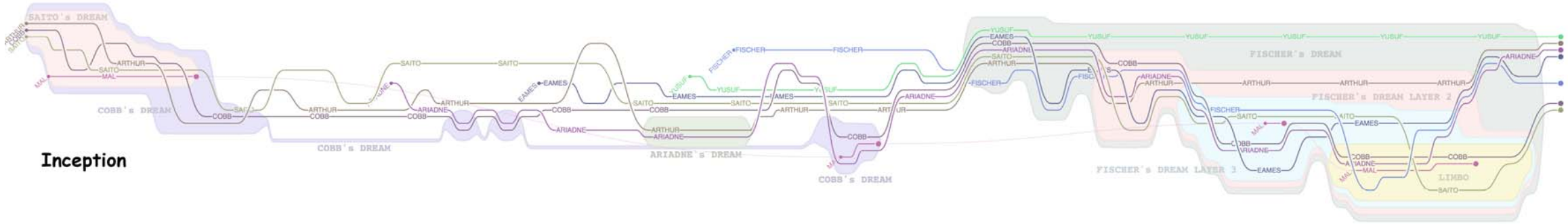
Storyline visualizations

- From [Tanahashi and Ma 2012]

i_k	t_k	d_k	M_k
i_1	0	4	SISTER, ALICE
i_2	4	0	WHITE RABBIT
i_3	5	5	ALICE,WHITE RABBIT
i_4	5	217	SISTER
i_5	11	21	WHITE RABBIT
i_6	11	6	ALICE







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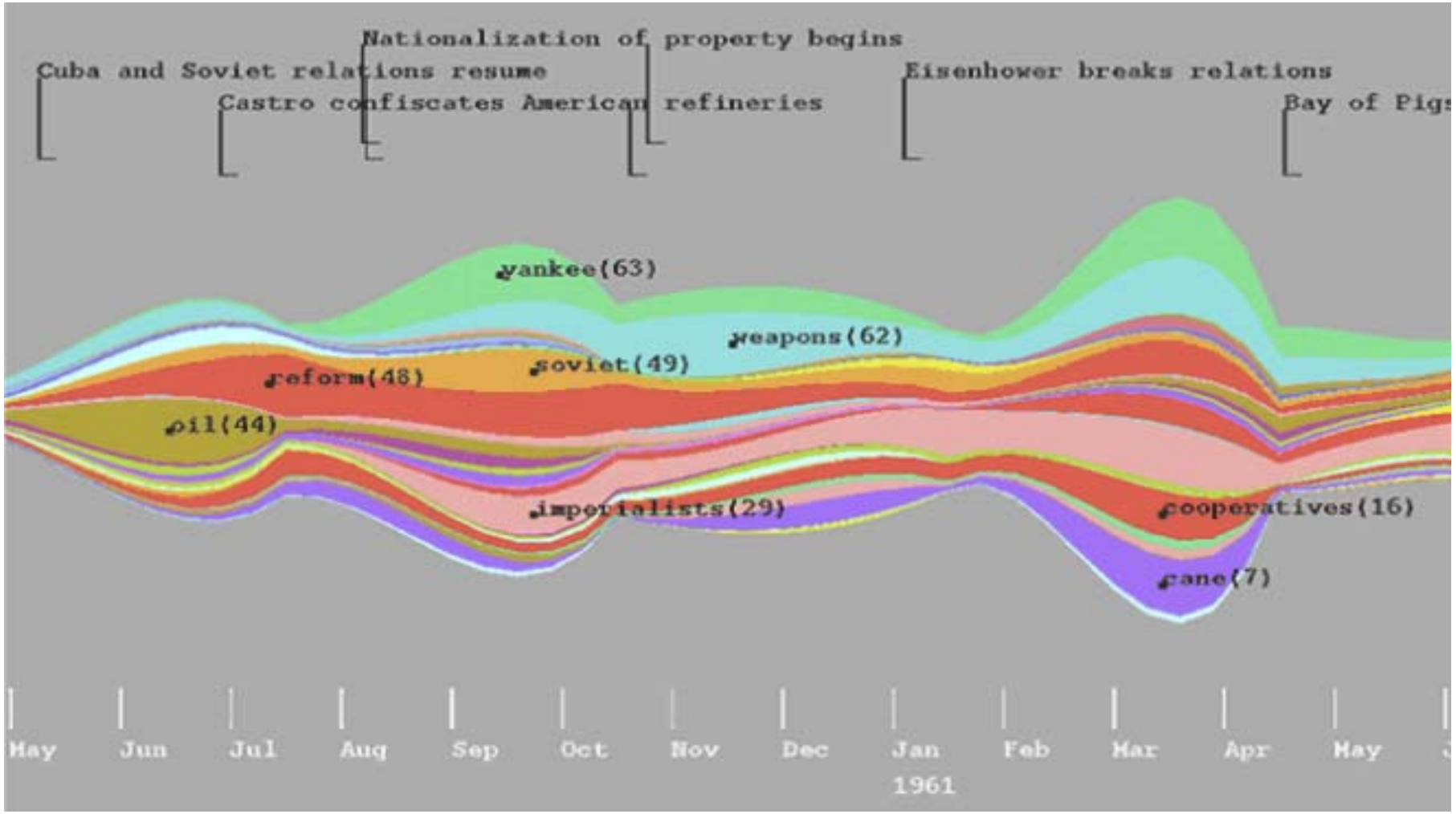
Case 3: Streams of text

- Consider a set of speeches, messages, or documents over time
- Can you represent the flow of ideas and concepts in such a collection?

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ThemeRiver

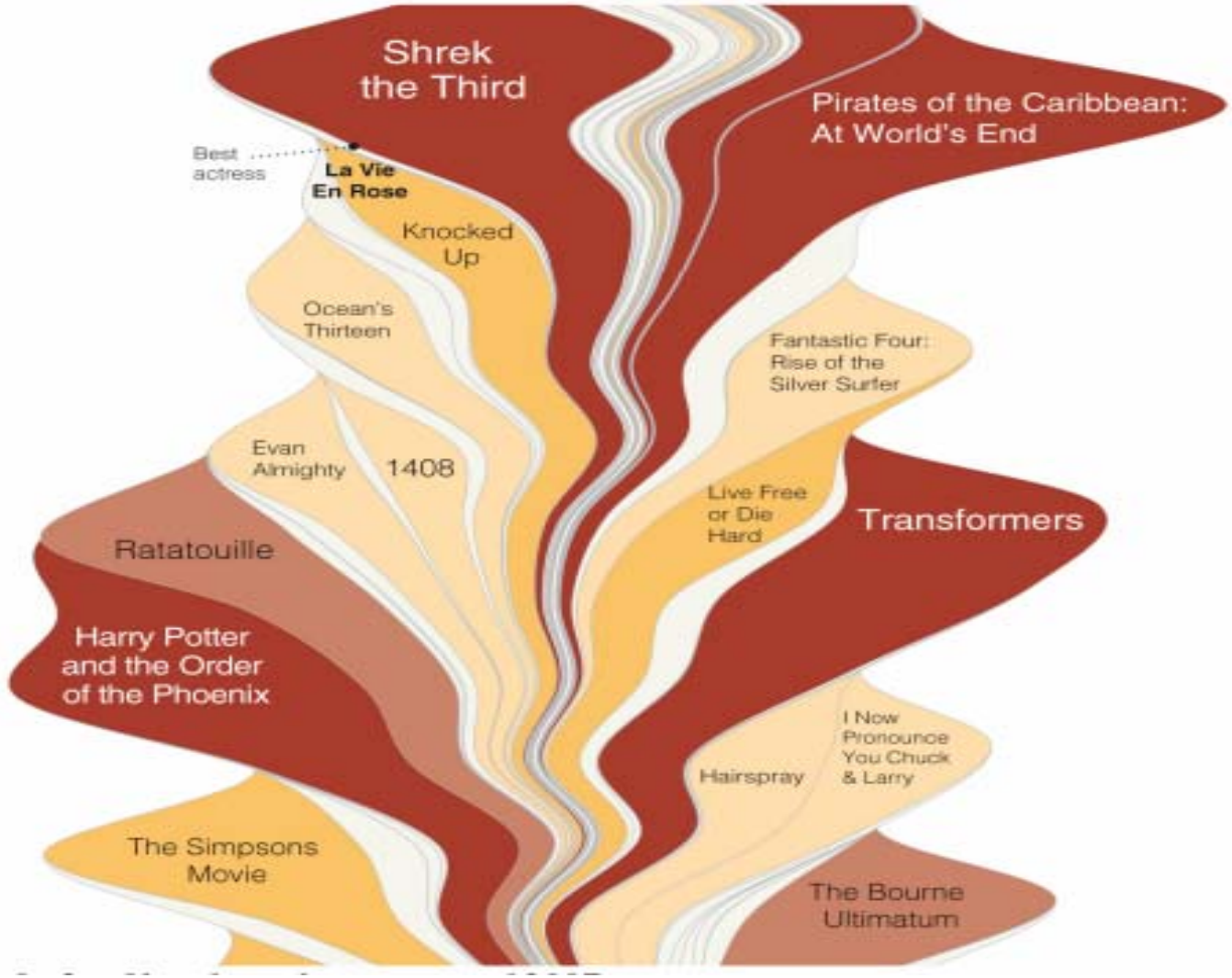
- [Havre *et al.*, 00]



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Stacked graphs

- From [Byron & Wattenberg, ‘08]
- Goals
  - show multiple time series
  - be able to see sum
  - make labels legible



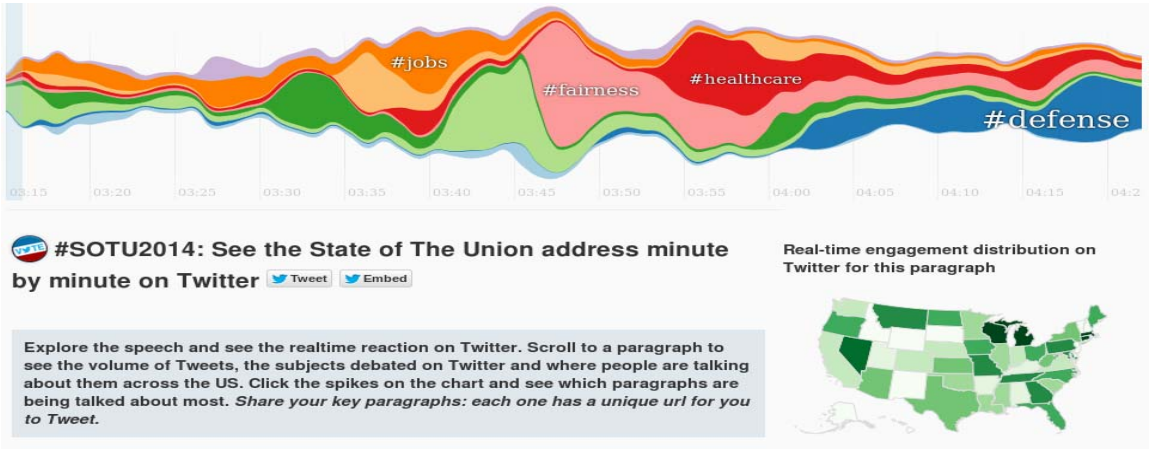
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Design issues

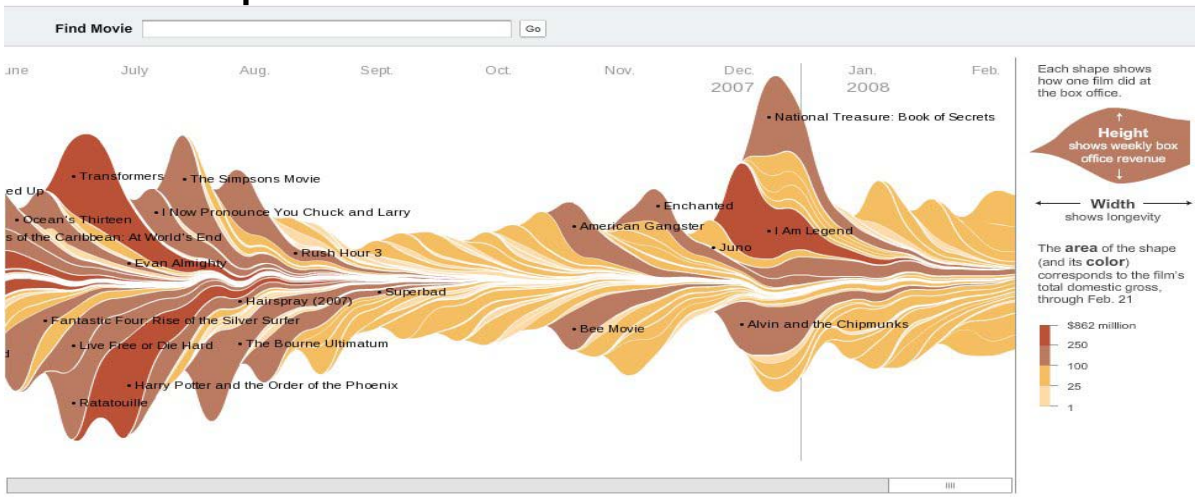
- Curve shape
  - Wiggle, symmetry, balance
    - Definitely some interesting math to do it
- Color choice
- Labeling
- Layer ordering

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ThemeRivers online



http://twitter.github.io/interactive/sotu2014/#p1



http://www.nytimes.com/interactive/2008/02/23/movies/20080223_REVENUE_GRAPHIC.html

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